Fish and Fisheries

WILEY

FISH and FISHERIES

ORIGINAL ARTICLE

Rebuilding and Reference Points Under Compensatory and Depensatory Recruitment: A Meta-Analysis of Northeast Atlantic Fish Stocks

Christoffer Moesgaard Albertsen[1] | Tommi Perälä[2] | Massimiliano Cardinale[3] | Henning Winker[3] | Vanessa Trijoulet[1]

[1]National Institute of Aquatic Resources, Technical University of Denmark, Kongens Lyngby, Denmark | [2]Department of Biological and Environmental Science, University of Jyväskylä, Jyväskylä, Finland | [3]Department of Aquatic Resources, Institute of Marine Research, Swedish University of Agricultural Science, Lysekil, Sweden

Correspondence: Christoffer Moesgaard Albertsen (cmoe@aqua.dtu.dk)

Received: 2 June 2024 | **Revised:** 29 October 2024 | **Accepted:** 4 November 2024

Funding: This work was supported by the European Maritime and Fisheries Foundation and the Ministry of Environment and Food of Denmark for the project 'Rebuilding Western Baltic Spring spawning herring' (33113-B-20-174), and the Academy of Finland (340901), In addition, this project has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (770884).

Keywords: Allee effect | depensation | errors-in-variables | meta-analysis | stock assessment | stock–recruitment relationship

ABSTRACT

Modern management of fish stocks is based on integrating the precautionary approach with the maximum sustainable yield framework. It relies on accurate estimation of precautionary limits, defined as levels of spawning biomass where a stock has reduced reproductive capacity, and harvesting targets aimed to maximise future yields. Therefore, it is heavily depending on productivity assumptions. Most fish stocks are managed assuming that productivity will increase as the stock size decreases (i.e., density dependent compensatory stock and recruitment relationship). However, several biological and ecological processes will result in a decreased productivity below a certain population size, referred to as the Allee effect or depensation. Through a meta-analysis of 81 Northeast Atlantic fish stocks, we investigated the impact of assuming compensatory recruitment in the presence of depensation in fisheries management. Across life histories, depensation results in a 22% reduction of the fishing mortality rate leading to extinction. On average, the maximum reproductive rate per spawning biomass was found at 35% of B_{MSY}, which was also the biomass where stocks have a 5% risk of extinction without fishing. Finally, the presence of depensation resulted in increased rebuilding times when stock spawning biomass falls below the limit reference point. When depensatory effects are present, assuming increasing productivity at low biomass will generally result in over-optimistic perceptions of rebuilding and stock status at biomass below 25% and 45% of B_{MSY} in general, and for pelagic stocks respectively. When not accounted for, depensation will potentially lead to unsustainable harvesting practices of marine living resources.

1 | Introduction

Sustainable utilisation of marine living resources depends on setting precautionary limits to avoid overfishing. Limits on the minimum viable population size, defined as the level of spawning biomass below which a stock is considered to have reduced reproductive capacity (i.e., limit reference point), are functions of both life-history parameters and the stock–recruitment relationship. Throughout the world, most stock–recruitment relationships used in fisheries management are assumed to have compensatory mortality property, where the number of recruits per spawner biomass is assumed to increase with decreasing stock size (Albertsen and Trijoulet 2020; ICES 2021a; Kolody et al. 2019; e.g., Methot and Wetzel 2013; Nielsen and Berg 2014; Sharma et al. 2019). For models that assume compensatory recruitment mortality, stocks will, on average, be predicted to recover from overfishing at any positive stock size. In contrast, models with depensatory mortality—or 'Allee effect'— (Courchamp, Berec, and Gascoigne 2008) will have decreasing recruitment rates below a certain stock size. As a result, recovery from overfishing will be impaired or even impossible.

Depensatory effects can be explained by different ecological mechanisms such as the difficulty to find mates, the difficulty to avoid predation (e.g., for schooling fish) or the decrease in foraging success (Kramer et al. 2009). There is also evidence that selective fishing can shift the balance in top-down controls among species, thereby causing depensatory mortality through increased predation on early life stages (e.g., Worm and Myers 2003). While biological principles suggest that population dynamics become depensatory at extremely low abundances (consider the extreme case of a population consisting of only one individual), the precise threshold at which per capita recruitment becomes impaired, and the point at which stock collapse occurs, remain uncertain. Equally uncertain is whether intensive fishing of commercially exploited fish stocks can ever lead to the crossing of this threshold, and therefore trigger the Allee effect. The adequacy of current management practices in averting this scenario is also an open question. Regrettably, the possibility of depensatory recruitment dynamics remains a seldom-addressed aspect in contemporary fisheries management.

In general, the evidence supporting depensation in marine fish stocks has been limited (Hutchings 2014; Liermann and Hilborn 1997; Myers et al. 1995). One primary reason for the scarcity of direct measurements of the Allee effect is the practical challenge of gathering data from small- or low-density populations, which are necessary to observe and quantify these effects (Gilroy et al. 2012). In marine systems, recruitment and spawning stock size are seldom directly observed, but instead derived as outputs from fisheries stock assessments, which are often associated with large uncertainty. Further, the complexity of environmental processes encountered during the early life stages of fish from egg to recruitment can mask depensatory effects. For example, difficulty to find mates may reduce the number of viable eggs in the same year as favourable conditions reduce the mortality of larvae and juveniles, which will thus mask a potential Allee effect. Likewise, assessing several spawning populations as one stock (Frank and Brickman 2000) as well as methodological issues may also pose additional obstacles in accurately assessing the presence of the Allee effect (Perala and Kuparinen 2017). Several northwest Atlantic cod (*Gadus morhua*, Gadidae) stocks have collapsed in the 1990s and have not recovered despite a moratorium on fishing. While many studies argue that the absence of recovery is mostly due to predation by grey seals (*Halichoerus grypus*, Phocidae, e.g., Benoît et al. 2011; Trzcinski, Mohn, and Bowen 2006) or climate change (e.g., Sguotti et al. 2019), depensation was also proposed as potential reason for some stocks (Neuenhoff et al. 2019; Perala, Hutchings, and Kuparinen 2022; Swain and Benoît 2015).

Some assessments and short-term forecasts bypass the assumption of the functional form of the relation between stock and recruitment by using a constant level of recruitment with deviations (e.g., Needle 2001) or a time-series model on recruitment (e.g., Nielsen and Berg 2014). However, optimal harvest control rules for rebuilding stocks in the medium term as well as precautionary limit reference points are highly dependent on stock–recruitment assumptions. Several jurisdictions aim for fishing at levels producing the maximum sustainable yield (MSY, UN (United Nations) 1995), where MSY is highly dependent on the assumed recruitment dynamics (Albertsen and Trijoulet 2020; e.g., Barrowman and Myers 2000). Likewise, limit reference points are often determined as a fraction of B_{MSY}, the equilibrium biomass when fishing at a level producing MSY, or a fraction of the unfished equilibrium biomass, B_0 (González Herraiz et al. 2023; ICCAT (International Commission for the Conservation of Atlantic Tunas) 2019; ICES 2024; New Zealand Ministry of Fisheries 2011; NRC (National Research Council) 2014). When estimating MSY reference points using a stock–recruitment relationship (i.e., ignoring proxy MSY reference points), most jurisdictions use models that are compensatory and assume increasing recruitment rates at low stock abundance (ICES 2021a; e.g., segmented regression or hockey stick, Ricker, Beverton–Holt, New Zealand Ministry of Fisheries 2011; NRC (National Research Council) 2014).

While using compensatory models is typically the default, several stock–recruitment models have been proposed to account for possible depensatory mechanisms of recruitment. Those models are either derived by calculating the probability of mating (Hilborn et al. 2014; Liermann and Hilborn 2001; e.g., McCarthy 1997), or by adding a parameter to a classic compensatory recruitment model. The sigmoidal Beverton–Holt (Myers et al. 1995) and the Saila–Lorda (e.g., Iles 1994; Needle 2001) recruitment models add an exponent to the stock biomass of the Beverton–Holt and Ricker models respectively. When the exponent parameter is above one, the model is depensatory. Similarly, Frank and Brickman (2000) incorporated depensatory dynamics in a Ricker recruitment model by subtracting a depensation parameter from the spawning biomass. In contrast, Barrowman et al. (2003) incorporated depensation by scaling the Beverton–Holt and logistic hockey stick recruitment curves with a sigmoidal function of spawning biomass.

Using a comprehensive dataset of 81 Northeast Atlantic fish stocks, we considered the potential consequences of ignoring depensation in recruitment, when it is detectable. These consequences are evaluated through a meta-analysis, and synthesised across life histories and ecosystems. The analysis compares reference points and rebuilding potential between compensatory and depensatory models. For each stock, we fit one compensatory and four depensatory versions of four different recruitment models: the Beverton–Holt, Ricker, Hassell/Deriso and hockey stick. To construct depensatory versions of the four recruitment curves, we use the approaches presented above and propose an additional two-parameter sigmoidal scaling model.

2 | Literature Synthesis

Before turning to meta-analysis, we provide a concise, structured synthesis of the relevant literature. The synthesis is based on a query to the Web of Science Core Collection. Further, we provide a detailed synthesis of depensatory modifications in Section 3.2.2 'Depensation types' below.

2.1 | Scoping of the Synthesis

To synthesise the literature on depensatory recruitment in fish stocks, the Web of Science Core Collection was searched for articles containing 'depensation', 'depensatory' or 'Allee' and either 'recruitment' or 'recruit' as well as 'fisheries', 'fishery', 'fishing' or 'stock assessment' in the title, abstract, author keywords or the Web of Science defined 'keywords plus'. Articles were manually screened to exclude review papers, perspective papers and false positives that did not model depensatory recruitment.

Manuscripts were classified by study type (data fitting or theoretical/simulation) and modelled dynamics (biomass, age-based, stage-based, length-based, empirical). For models including a stock–recruitment model, the type of depensation was recorded. Further, evidence of the presence of depensation was recorded along with effects on target reference points, limit reference points, rebuilding and harvesting. Finally, the data source and model species were registered.

2.2 | Characteristics of the Literature

The query returned 121 articles (accessed 7 August 2024). Of these, 58 were excluded: two were false positives (e.g., matching 'non-depensatory'), 11 were perspective or review papers and 45 were excluded because they only discussed depensation or did not include depensatory recruitment. Further, 20 papers were borderline relevant. These included papers that investigated depensatory recruitment without an explicit recruitment model or accounted for depensation but without presenting a comparison to a case without. The remaining 41 papers matched the inclusion criteria.

Of these included studies, five were meta-analyses, 23 were predominantly data based and 13 were predominantly theoretical or simulation studies. All five meta-analyses and 16 of the data studies were based on assessment model output. None of these accounted for the potential bias from using model estimates as observed data (e.g., Brooks and Deroba 2015; Dickey-Collas et al. 2014).

2.3 | Depensatory Recruitment Models

Several models have been used to model depensatory recruitment. The majority of studies used depensatory modifications of compensatory models. We refer to Section 3.2.2 'Depensation types' below for further details on functional forms. A power function of the spawning stock biomass (SSB) entering the stock–recruitment relationship (i.e., Type A depensation below) was used in 20 of the 41 studies, predominantly the sigmoidal Beverton–Holt (proposed by Myers et al. 1995) and Saila-Lorda models (e.g., Iles 1994). Six studies used the spawning stock biomass offset model (Type E below) proposed by Frank and Brickman (2000), five studies used the mating probability model (Type C below) proposed by Hilborn et al. (2014) and three studies used the sigmoidal scaling (Type B below) of Barrowman et al. (2003). Two papers used unique modifications (Todd and Lintermans 2015; Winter, Richter, and Eikeset 2020). Apart from functional forms of depensatory modifications, three studies used semiparametric recruitment models that can allow for depensation (Maroto and Moran 2014; Sugeno and Munch 2013a, 2013b), two studies constructed recruitment equilibria (A. Solari, MartinGonzalez, and Bas 1997; Solari et al. 2010) and two studies used empirical methods (Keith and Hutchings 2012; Zebro et al. 2022). Finally, Vergnon, Shin, and Cury (2008) included depensation indirectly via predation, and Prince (2013) considered a discontinuous cutoff model where recruitment was zero below a threshold and followed a Beverton–Holt model above, which results in a jump in recruitment at the threshold. Only four of the 41 studies included more than one depensatory type. No study included more than two depensatory forms. It was more common to include more than one compensatory model, which was typically modified by the same depensatory type. This was done in 13 of the studies.

2.4 | Evidence for Depensation

Ranging from single-stock studies to large meta-analyses, 25 of the 41 studies looked for evidence of depensation. In a Bayesian meta-analysis of 82 walleye stocks, Sass, Feiner, and Shaw (2021) found that around half of the stocks had depensatory posterior modes, although most credible intervals overlapped with 0. Further, most of the slope estimates near zero indicated a weak stock–recruitment relationship. The results were corroborated by a study on Walleye across 20 lakes, which found increased age-0 mortality at low catch per unit effort (Zebro et al. 2022).

The three other meta-analyses were based on the RAM Legacy Stock Assessment Database (RAM Legacy Stock Assessment Database 2024). Across 207 stocks from 104 species, Keith and Hutchings (2012) looked at standardised recruits-per-spawner grouped in 10% fractions of maximum observed SSB. In general, recruits-per-spawner tended to increase for lower SSB, but with a slowing increase at the lowest SSB level. This indicates a general decline in the strength of compensation, which cannot be captured by, for example, the Ricker and Beverton–Holt recruitment models. For more than 20% of the stocks, they found a decline in recruits-per-spawner at the lowest observed SSB level, indicating depensation. However, in most cases, the difference from 0 was not statistically significant. A strong Allee effect was found for cod stocks, mainly driven by western Atlantic stocks. In a Bayesian analysis of 17 cod and flatfish stocks, including a change-point analysis, Tirronen, Perala, and Kuparinen (2022) found evidence of depensation in four flatfish stocks. Each had at least one regime with more than 80% posterior probability of depensation. The analysis included depensation using the sigmoidal Beverton–Holt and the Saila–Lorda recruitment models (i.e., Type A depensation below).

In another Bayesian meta-analysis of 114 stocks using the sigmoidal Beverton–Holt recruitment model, M. Liermann and Hilborn (1997) also found that the posterior mode of the depensatory parameter indicated depensation for flatfishes. However, for three other taxa (Gadiformes, Salmonids and Clupeiformes), the posterior mode indicated hyper-compensation. In all cases, posterior distributions were wide with broad tails covering the entire valid range. Later, Hilborn et al. (2014) analysed 113 stocks in a meta-analysis. In this analysis, depensation was included by scaling the SSB for a Deriso recruitment model with the probability of finding a mate (i.e., Type C depensation below). Looking at individual stocks, only four showed an improved AICc when depensation was included in the Deriso model. The results are similar to Myers et al. (1995), which was not included in the query results from Web of Science, who found significant depensation in 3 of 128 stocks using a sigmoidal Beverton–Holt model (i.e., Type A depensation below).

In studies of individual stocks, Hemeon et al. (2020) could not fit recruitment models without including depensation for Eastern oyster (Type E depensation below), and Marks et al. (2021) found that including depensation (Type C depensation below) improved their biomass model for blue swimmer crab. In contrast, Bechtol and Kruse (2009) did not find statistically significant depensation for red king crab.

Nine studies analysed individual cod stocks using either stock–recruitment pairs, age-structured models or biomass models. Depensation was found in six of the studies (Cabral, Alino, and Lim 2013; Li and Yakubu 2012; Maroto and Moran 2014; Perala, Hutchings, and Kuparinen 2022; Sugeno and Munch 2013a; Winter, Vasilyeva, and Vladimirov 2023). A. Solari, MartinGonzalez, and Bas (1997) and Marshall et al. (2006) found weak indications of depensation, while Tirronen, Depestele, and Kuparinen (2023) found a stronger effect of environmental variables. Tirronen, Depestele, and Kuparinen (2023) also included two flatfish stocks. Likewise, Yakubu et al. (2011) found no evidence of depensation in a cod stock, while depensation provided a better fit for a halibut stock.

For salmonids, evidence of depensation was limited. In a Pacific salmon stock, Quinn et al. (2014) found evidence of increased predation mortality at low abundance, indicating Allee effects, but depensation was not detectable in the stock–recruitment relation. Likewise, Gurney et al. (2010) found that there was no evidence of depensation in the complete stock–recruitment relation; however, some life-cycle stages did indicate depensation. Further, Chen, Irvine, and Cass (2002) found no indication of depensation for a sockeye salmon stock, but significant depensation for a coho salmon stock. However, studying 14 coho salmon stocks, Barrowman et al. (2003) did not find evidence of depensation.

In studies of pelagic stocks, evidence of depensation was also variable. Morales-Bojórquez and Nevárez-Martínez (2005) studied pacific sardine and found indications of depensation. However, the stock had not been in the critical biomass region. In a total of 12 Northeast- and one Northwest Atlantic herring stocks, evidence of depensation was found in seven (Nash, Dickey-Collas, and Kell 2009; Perala and Kuparinen 2017; Sugeno and Munch 2013b). However, using a Gaussian process to model recruitment, Sugeno and Munch (2013b) found an 85% risk of not detecting mild depensation and 58%–73% risk of not detecting strong depensation depending on the model, which highlights that a lack of evidence for depensation in marine species cannot in general be considered evidence of a lack of depensation.

2.5 | Effect on MSY Reference Points

One of the 41 studies that matched the inclusion criteria directly considered the effect of depensation on MSY reference points. In their biomass model for blue swimmer crab, Marks et al. (2021) found that, in theory, depensation will increase B_{MSY} if all other parameters remain the same. However, estimated B_{MSY} was reduced by 40% for the Schaefer model and increased by 10% for the Fox model. MSY increased substantially in both cases, although the estimates were very low for the models without depensation. Further, in an artificial fertilisation experiment on rainbow smelt, Purchase, Hasselman, and Weir (2007) (classified as borderline relevant by our selection criteria) found that the mean per cent of eggs fertilised increased with the number of males contributing to the fertilisation while the variability decreased. Compared to a 100% fertilisation rate, the reduction for five donors corresponded to an 18.6% reduction in F_{MSY}, while the reduction was around 50% for one donor.

2.6 | Effect on Harvesting

The impact on harvesting with depensatory recruitment was investigated in nine of the included studies: one data-based study and nine theoretical or simulation studies. Fitting a depensatory (Ricker Type A below) recruitment model North Sea cod assessment outputs, Cabral, Alino, and Lim (2013) forecasts the population in a status quo fishing scenario and estimates the fishing mortality rate that would lead to extinction. In one case, the status quo is above that level although they did not compare to a compensatory model.

In sedentary species, spatial depletion may lead to local Allee effect. Gonzalez-Duran et al. (2018) investigated the impacts of the Allee effect in a theoretical, spatial bioeconomic model inspired by sea cucumber (combining Type C and Type E depensation below). While the open-access fishery reached the bioeconomic equilibrium when there was no Allee effect, the stock collapsed in 15 years when depensation was considered. In that case, the fishery altered the spatial distribution of the sedentary stock, thereby invoking depensation in recruits per spawner. Accounting for depensation, the optimum effort was 150 boats per year compared to 691 under the open-access compensatory scenario. Not accounting for the Allee effect would lead to an estimated loss of USD 31.8 million over 25 years. Likewise, in a theoretical study based on Queen conch, Chan and Kim (2014) constructed a spatial stage-structured biomass model with sigmoidal Beverton–Holt recruitment (Type A depensation below). They found that maintaining a marine reserve is favourable in terms of both abundance and yield. Likewise, Quinn, Wing, and Botsford (1993) found that harvest refugia can provide protection against catastrophic collapse of fisheries on red sea urchin, if effort is difficult to control or Allee effect thresholds are difficult to estimate. These results were corroborated by Pfister and Bradbury (1996) (categorised as borderline relevant by our selection criteria), who found that very low fishing rates and rotation schedules of at least 2 years were needed for the red sea urchins to persist over the 100-year simulations.

In nonspatial models, several studies also found a higher risk of stock collapse under depensatory recruitment at the same exploitation levels (Golden, van Poorten, and Jensen 2022; Shertzer and Prager 2007; Vasconcellos 2003). However, the risk may be mitigated by an effective management response with feedback control (e.g., Shertzer and Prager 2007).

Overall, the consensus is that harvesting should be more precautionary in the presence of the Allee effect, since the risk of extinction is greatly increased. The risk can be reduced through threshold harvest control rules or area closures. However, since most of the studies are theoretical, the effects are often evaluated keeping other parameters constant. When estimating parameters from data, all parameters will be influenced by including depensation in the recruitment model. The influence of the combination of these effects remains an open question.

2.7 | Effect on Rebuilding

Depensation in recruitment lowers the number of recruits per spawner at low stock abundance. Consequently, a longer rebuilding time of depleted stocks is expected. Five of the 41 studies investigated the influence of depensation on rebuilding. In a theoretical study of an 'average fished population', Shertzer and Prager (2007) found that rebuilding time was increased when including depensation. In some scenarios, rebuilding time was doubled compared to the similar compensatory scenario. Focusing on North Sea cod, Winter, Richter, and Eikeset (2020) also found that depensation prolongs rebuilding and increases the risk of collapse. The effects can be worsened by both ineffective management interventions and increasing sea surface temperatures. In another study of Atlantic cod, covering 17 stocks, Winter, Vasilyeva, and Vladimirov (2023) also find that stocks with an Allee effect show a longer recovery time and an increased risk of collapse. The effects were worsened by increase in temperature. However, they did not find a correlation between the strength of the Allee effect and recovery time.

Finally, another two studies investigated the impact on recovery from reseeding and translocating of wild stocks. In a cost–benefit analysis of natural recovery versus reseeding and translocating for rebuilding abalone populations, Prince (2013) found that in the absence of depensation, natural rebuilding by reducing fishing and rapidly boosting spawning biomass had the greatest cost–benefit. In contrast, when population dynamics were depensatory, the stock did not rebuild naturally, and translocation of adults had the greatest cost–benefit. However, in a simulation study of perch, Todd and Lintermans (2015) found that increasing the strength of depensation will reduce the effectiveness of translocation, thereby requiring translocation of more individuals to ensure a successful strategy.

While the consensus from these studies is that depensation will increase rebuilding time and the risk of stock collapse, studies on marine fishes have been limited to Atlantic cod. Further, the influence will depend on the level of depletion. However, the depletion level where a difference occurs between assumed compensation and depensation dynamic, inferred from data, was not investigated in any of the studies.

2.8 | Knowledge Gaps

Many studies have investigated the Allee effect in marine fishes. In this synthesis of relevant literature, we found that data-based studies, including several large meta-analyses, predominantly focused on the presence of the Allee effect in marine fishes. Further, theoretical studies have investigated the impacts of the Allee effect on harvesting and rebuilding, but since these studies were not based on data, they typically stipulated a depensation parameter value without any change of other parameters. However, when estimating parameters from data, other parameters would also be influenced by the inclusion of depensation in recruitment. Finally, the Web of Science Core Collection query only returned one paper which compared MSY reference points between compensatory and depensatory models for biomass dynamic models. None of the papers explicitly compared the impacts of depensation in recruitment on MSY reference points for marine fishes.

Through this synthesis of previous studies, we have identified the three main gaps in the current literature that we seek to fill with this meta-analysis. Firstly, previous meta-analyses have solely focused on statistical significance of depensation parameters in marine fishes. Structurally, statistically significant depensation is difficult to find, since stock–recruitment 'data' in marine fishes are often indirect, noisy assessment output, time series are short and the span of observed biomass levels can be narrow. From first principles, we conclude that depensation must happen in most, if not all, species at some biomass level. Therefore, our approach is a meta-analysis of the difference between the perception of stocks assuming compensatory and depensatory recruitment, respectively, in terms of MSY reference points, rebuilding and Allee effect thresholds. As a result, we synthesise the potential risk from assuming compensation in fisheries stock assessments if the true system is depensatory at the level inferred from the data. The data used in our analysis span several ecosystems, environmental regimes, fish species, life histories and fisheries.

Secondly, most of the studies listed in the synthesis above include only one type of depensation in the recruitment model. If the functional form of depensation does not sufficiently match the true system, there is a risk of concluding that compensatory recruitment is a better fit. To limit the effect of imposing a restrictive functional form, we considered four depensatory modifications of each of the four different compensatory recruitment models. However, to restrict our study to the effect of depensation, we identified the best-fitting model in terms of AIC and compared the corresponding compensatory model to the best-fitting depensatory modification of that recruitment model.

Finally, the synthesis included 22 studies (and two classified as borderline relevant by our selection criteria) that were based on modelling stock–recruitment pairs estimated from an assessment model. None of these accounted for the fact that both the estimated stock size and recruitment come with estimation uncertainty (e.g., Brooks and Deroba 2015; Dickey-Collas et al. 2014). Typically, uncertainty is not recorded in assessment output databases (such as the RAM Legacy Stock Assessment Database RAM Legacy Stock Assessment Database 2024). However, by combining several databases including stocks assessed by the International Council for the Exploration of the Sea (ICES), we include estimation standard errors to the stock–recruitment pairs used in the meta-analysis. As a result, we can account for estimation uncertainty in the stock–recruitment pairs when fitting recruitment curves using an errors-in-variables approach. While this analysis is restricted to the Northeast Atlantic (FAO Major Fishing Area 27) and two Greenlandic cod stocks (FAO Major Fishing Area 21) by data availability, the approach can be applied without modification if similar data from other regions become available.

3 | Methods

To investigate the potential consequences of ignoring depensation in recruitment, a simulation study was conducted based on the life histories of 81 Northeast Atlantic fish stocks (two are from west of Greenland, which is Northwest Atlantic, ICES 2021b, 2022a). The stocks covered a range of ecosystems, habitats and species (Figure 1). For each stock, a compensatory and four different depensatory models combined versions of each of the four recruitment models were fitted to the dataset. Based on the estimated stock–recruitment models, stock trajectories were simulated forward from different starting spawning stock biomass levels to evaluate the impact of depensation on reference points, rebuilding time and rebuilding potential.

3.1 | Database

The original dataset used in this study comprises stock assessment inputs and outputs for, originally, 81 stocks from the entire Northeast Atlantic ICES region. The dataset was collated in the form of stock objects of the FLStock class as defined in the FLR (Kell et al. 2007) framework (Griffiths et al. 2024; Table S1, ICES 2022a). The 81 stocks are all classified by ICES as Category 1 (i.e., age-structured analytical stock assessment) and were last assessed by ICES in 2019 ($n=12$), 2020 ($n=63$) or 2021 ($n=6$). In the following text, stocks are referred to by ICES stock IDs; details on the assessment input and outputs of all stocks are provided in the form of an external open-access Shiny Application (https://maxcardinale.shinyapps.io/Indicators/, ICES 2022a). The Shiny App also contains a range of plots that visualise various aspects of each stock's population dynamics and demographic characteristics.

A graphical summary of the dataset is provided in Figure 2. The dataset includes assessment inputs and outputs from 11 different age-structured modelling platforms, of which SAM ($n=36$, Nielsen and Berg 2014) and Stock synthesis ($n=14$, Methot and Wetzel 2013) are the most common (see ICES 2022a for more details). The 81 stocks comprise 25 bony fish species (representative of nine taxonomic orders) as well as one crustacean, *Pandalus borealis*. For this meta-analysis, two stocks were excluded, thus a total of 79 stocks were used for the analysis. The

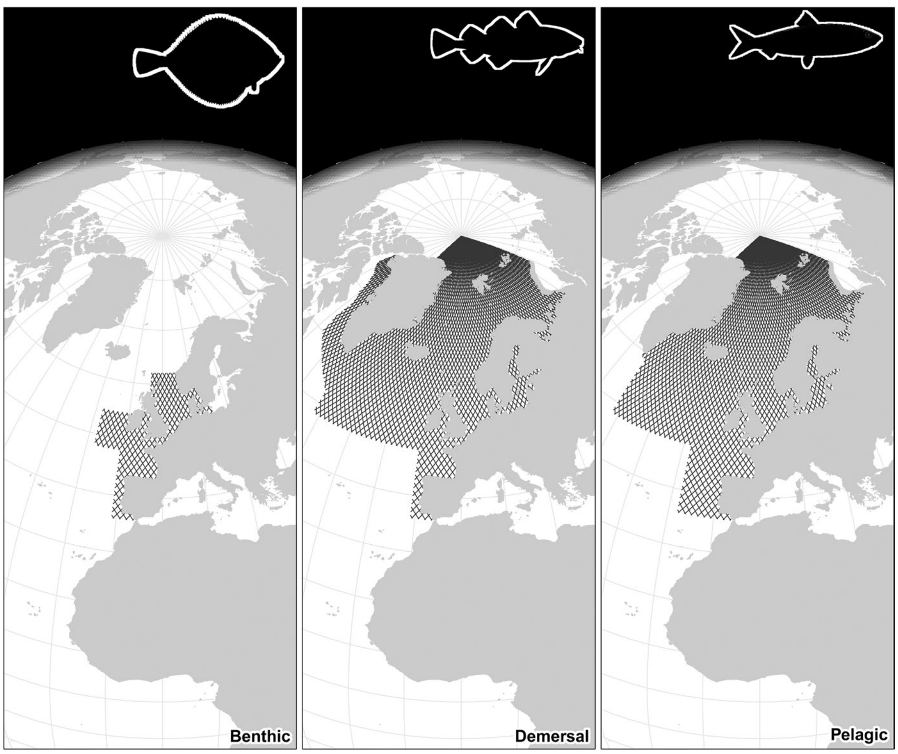

FIGURE 1 | Spatial coverage of the stocks included in the meta-analysis by fisheries guild (see Table S1). Fish icons are adapted from The New York Public Library, https://digitalcollections.nypl.org/collections/ichtylogie-ou-histoire-naturelle-gnerale-et-particulire-des-poissons#/?tab=navigation.

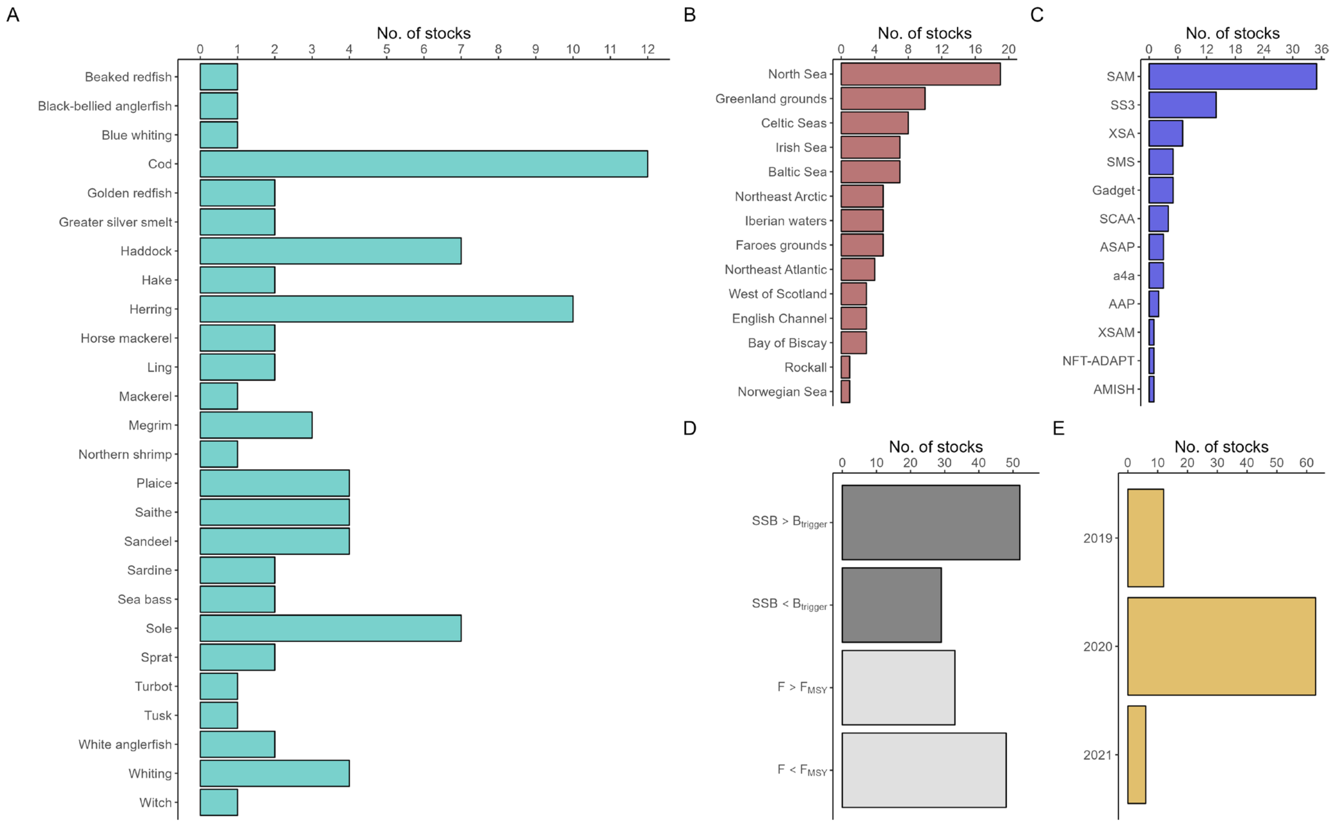

FIGURE 2 | Graphical summary of the dataset used in this study. The dataset contains 26 species (A) from 14 different areas (B), whereby area is a broad characterisation based on ICES areas and ecoregions, with different stock status (D), across different years (E). All 81 stocks are assessed using age-structured analytical assessment frameworks (C), the details of which can be found in Table S1. Reproduced from Griffiths et al. (2024).

two stocks were the single stock from the crustacean fisheries guild (Northern shrimp in the North Sea; i.e., pra.27.3a4a), and a cod stock, which was excluded because the depensatory fits had estimated equilibrium unfished biomass of zero, indicating that the estimated recruitment relation is not sufficient to sustain the stock with the current levels of natural mortality (i.e., Eastern Baltic cod; i.e., cod.27.24-32). Taxonomic information for each species was sourced from FishBase (Froese and Pauly 2023) through the rfishbase R package (version 4.1.1, Boettiger, Temple Lang, and Wainwright 2012). Further, stocks were grouped by fisheries guild as defined by ICES (benthic, crustaceans, demersal or pelagic, available from ICES 2023c). The fisheries guild combines biological and fisheries properties: benthic stocks are bottom-resting fish primarily caught by bottom contact gears (e.g., flatfish and monkfishes); demersal stocks are other fish mainly caught by bottom contact gears (e.g., most cod-like fish and sandeel) and pelagic stocks are mainly caught by midwater gears (e.g., herring, sprat and mackerel). Most stocks belong to the following three taxonomic orders Gadiformes ($n=33$, one pelagic and 32 demersal), Pleuronectiformes ($n=16$, all benthic) and Clupeiformes ($n=14$, all pelagic). Note that there is only one Chondrichthyes species (Northeast Atlantic spurdog) that is assessed as Category 1 by ICES; however, the assessment of this stock is not included in our database.

The stock–recruitment pairs from the assessments are model estimates. Therefore, the stock data collected by ICES (2022a) and Griffiths et al. (2024) were combined with the corresponding standard errors extracted from the ICES Stock Assessment Database (ICES 2023b) using the icesSAG R package (Millar et al. 2022, version 1.4.0). The ICES Stock Assessment Database contained confidence intervals on natural scale. Since both number of recruits and spawning biomass must be nonnegative and are, therefore, routinely estimated on log scale in assessment models, it was assumed that the log-transformed confidence intervals would be symmetric. Therefore, standard deviations were calculated by subtracting the estimates from the highest endpoints of the confidence interval on log scale and dividing by two. The two data sources were combined using ICES stock unique keys and assessment years via the ICES Stock Information Database (ICES 2023c). For stocks where corresponding standard errors could not be obtained, a common value calculated from all other stocks in the database was used. To calculate the proxy standard error, the average standard error over the years was calculated for each stock. In turn, the median of these was calculated and used as the proxy. Joining of the two databases was necessary to obtain the data for this study, since the ICES Stock Assessment Database contained stock–recruitment pairs and corresponding confidence intervals, and the database collected by ICES (2022a) and Griffiths et al. (2024) contained stock–recruitment pairs and life-history parameters by age.

3.2 | Stock–Recruitment Models

Different stock–recruitment functions were fitted to the stock–recruitment data. In this study, four common compensatory recruitment models were considered. Further, each of them was modified to be depensatory. The recruitment models are illustrated in Figure 3.

3.2.1 | Compensatory Recruitment Models

With a compensatory stock–recruitment model, recruitment R is modelled as a function of spawning stock biomass, S, such that $\frac{\partial}{\partial(S)}R(S)/S < 0$. That is, the recruitment rate is increasing as the stock size decreases.

3.2.1.1 | Beverton–Holt. The fist compensatory model considered was the Beverton–Holt model. It was parameterized by maximum recruitment, R_{max}, and the biomass that gives half of the maximum recruitment, S_{50}, such that:

$$R_{BH}(S) = \frac{R_{max}}{1 + \frac{S_{50}}{S}}.$$

3.2.1.2 | Ricker. The second recruitment model considered was the Ricker model. The model was parameterised by the maximum recruitment, R_{max}, and the biomass at which the maximum recruitment is obtained, S_{max}, such that:

$$R_{\text{Ricker}}(S) = R_{max}\frac{S}{S_{max}}exp\left(1 - \frac{S}{S_{max}}\right).$$

3.2.1.3 | Hassell/Deriso. In addition to the Beverton–Holt and Ricker model, the three-parameter model suggested by Hassell (1975) and Deriso (1980) was implemented. Similar to the Ricker model, it was parameterised by the maximum recruitment, R_{max}, and the corresponding biomass, S_{max}, such that:

$$R_{\text{HD}}(S) = R_{max}\frac{S}{S_{max}}\left(1+\gamma-\gamma\frac{S}{S_{max}}\right)^{\frac{1}{\gamma}}.$$

In this analysis, the shape parameter, γ, was constrained to be between −1 and 0. When the shape parameter, γ, approaches zero, the recruitment curve will converge to the Ricker model; and as the shape parameter approaches −1, the model will resemble the Beverton–Holt recruitment curve.

3.2.1.4 | Segmented-Regression (Or Hockey Stick). The final stock–recruitment model considered was a smoothed version of the segmented regression, or hockey stick, model (Barrowman and Myers 2000). Similar to the other recruitment models, this model was parameterised by maximum recruitment, R_{max}, and the lowest biomass where maximum recruitment occurs, S_{max}; that is, the breakpoint between the increasing and the flat segments. The model was implemented as:

$$R_{\text{SR}}(S) = \frac{R_{max}}{S_{max}}\min\left(S, S_{max}\right),$$

where the minimum function was calculated as:

$$\min(a,b) = exp\left(\frac{1}{2}(loga + logb - |loga - logb|)\right).$$

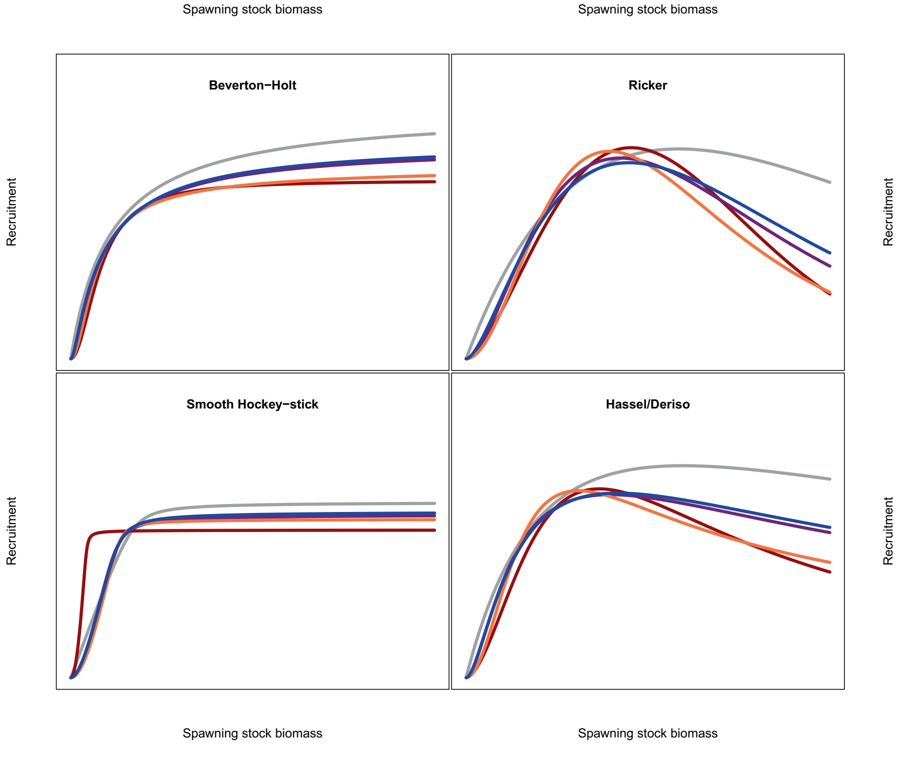

FIGURE 3 | Illustration of recruitment models included in the meta-analysis using an equally weighted average of parameters across stocks on the scale of estimation for the compensatory (grey), Type A (red), Type B (purple), Type C (orange) and Type D (blue) depensatory models. Since specific spawning stock biomass and recruitment values are not meaningful in this illustration, axes values have been omitted.

In turn, the absolute value was approximated by a smooth function:

$$|x| = \sqrt{x^2 + \epsilon}.$$

In this study, ϵ was fixed to 0.1. Lower values tended to give convergence issues while higher values gave smoother curves than intended.

When the hockey stick model is implemented with the true absolute value, it is not strictly compensatory. In this case, $\frac{\partial}{\partial S}\left(\frac{R_{SR}(S)}{S}\right) = 0$ when $S \leq S_{max}$, while the derivative is $-\frac{R_{max}}{S^2}$ when $S > S_{max}$. In contrast, the smoothed version implemented here is strictly compensatory. In this case:

$$\frac{\partial}{\partial S}\left(\frac{R_{SR}(S)}{S}\right) = \frac{-R_{max} \cdot exp(-\delta/2) \cdot (\delta + log(S) - log(S_{max}))}{2 \cdot S^{3/2} \cdot \delta} < 0,$$

where:

$$\delta = \sqrt{log(S)^2 - 2 \cdot log(S) \cdot log(S_{max}) + log(S_{max})^2 + \epsilon}.$$

3.2.2 | Depensation Types

Each of the compensatory models was fitted with four depensatory modifications. Here, the general forms of the depensatory modifications are described. The models fall into one of two overall categories. The first category scales the spawning biomass before applying the compensatory recruitment model, while the second scales the recruitment after applying the compensatory model to the full spawning biomass. We refer to Data S5 for specific details on each model implementation.

3.2.2.1 | Type A. The first form of depensatory modification, which we will refer to as type A, adds a parameter, d to the compensatory recruitment model, such that:

$$R_A(S, d) = R(S^d),$$

where $R(S)$ is the compensatory model. The type A model is depensatory for $d > 1$, compensatory for $d = 1$ and overcompensatory for $d < 1$. For the purpose of this analysis, the models were restricted to be depensatory with $d > 1$. There is no simple biological interpretation of the magnitude of the depensation parameter. Each of the models was re-parameterised to keep the interpretation of the original parameters. For the Beverton–Holt recruitment model, the Type A depensation is known as the sigmoidal Beverton–Holt model (Myers et al. 1995). Likewise, the Type A modified Ricker recruitment model (e.g., Cabral, Alino, and Lim 2013) is similar to the Saila–Lorda model (e.g., Iles 1994), with the key difference that not every S is exponentiated in the Saila–Lorda model.

3.2.2.2 | Type B. The second form considered (e.g., Barrowman et al. 2003), which we refer to as type B, multiplies the compensatory model by a sigmoidal function, $\frac{S}{S+d}$ with $d > 0$, such that:

$$R_B(S, d) = R(S)\frac{S}{S+d},$$

where $R(S)$ is the compensatory model. The depensation parameter was parameterised such that $d = d_0 \cdot S_{50}$ for the Beverton–Holt recruitment and $d = d_0 \cdot S_{max}$ for the other recruitment models. With this model, the depensation parameter can be interpreted as the size of the spawning stock (in the same unit as S) needed to reduce the expected number of recruits by 50% relative to a compensatory model with the same parameters (Barrowman et al. 2003). Each of the models were re-parameterised to keep the original interpretation of the parameters.

3.2.2.3 | Type C. The third depensatory form, referred to as type C, scales the spawning stock biomass by the probability of finding a mate (Hilborn et al. 2014):

$$R_C(S, d) = R\left(S \cdot \left(1 - exp\left(log(0.5)\frac{S}{d}\right)\right)\right).$$

The probability of mating was derived by Hilborn et al. (2014), assuming that individuals of the two sexes encounter each other at random. The depensation parameter, d, corresponds to the stock size that gives a 50% chance of mating. In the derivations by Hilborn et al. (2014), the probability of mating depends on the number of males. However, the assessments in this study did not have sex-specific data. Therefore, the probability is scaled by the spawning stock biomass instead. If the fraction of males is constant in the stock, the two formulations are equivalent. If the fraction of males varies, the lack of sex-specific information will add to the variability around the stock–recruitment curve. Depensatory modifications based on the probability of finding a mate with different assumptions were considered by, for example, McCarthy (1997) and Liermann and Hilborn (2001). Similar to the Type B depensation, the depensation parameter was parameterised as $d = d_0 \cdot S_{50}$ for the Beverton–Holt recruitment and $d = d_0 \cdot S_{max}$ for the other recruitment models. Each of the models was re-parameterised to keep the original interpretation of the parameters.

3.2.2.4 | Type D. The final depensatory form used here multiplies a two-parameter log-scale sigmoidal function to the compensatory recruitment model:

$$R_D(S, d_1, d_2) = R(S)\left(1 + exp\left(-d_2\left(logS - logd_1\right)\right)\right)^{-1}.$$

The first depensation parameter, $d_1 > 0$, is the biomass where the number of recruits is reduced by 50% relative to a compensatory model with the same parameters, similar to the depensation parameter in the Type B model. Again, the model was parameterised with $d_1 = d_0 \cdot S_{50}$ for the Beverton–Holt recruitment and $d_1 = d_0 \cdot S_{max}$ for the other recruitment models. The second depensation parameter, $d_2 > 0$, determines the slope of the sigmoidal function. As for the other model formulations, each of the models were re-parameterised to keep the original interpretation of the parameters.

3.2.2.5 | Other Depensatory Modifications. Besides the four models discussed above, Frank and Brickman (2000) proposed to model depensation by shifting the spawning stock biomass such that:

$$R_E(S, d) = \begin{cases} R(S-d), & S \geq d \\ 0, & S < d \end{cases}$$

In this model, the depensation parameter, $d > 0$, represents a biomass threshold for recruitment. This depensatory form was not included in this study, because it is not differentiable at d and the gradient with respect to S is zero below d. Therefore, it does not work well with the latent variable estimation framework used for the other models.

Another nondifferentiable modification was proposed by Winter, Richter, and Eikeset (2020). They scaled the recruitment model by an exponentially increasing function below a threshold:

$$R_F(S, d_1, d_2) = \begin{cases} R(S), & S \geq d_1 \\ R(S) \cdot exp\left(d_2 \cdot (S - d_1)\right), & S < d_1 \end{cases}$$

In this model, $d_1 > 0$ is a threshold below which the recruitment is affected and $d_2 > 0$ determines the severity of the depensatory effect. The model is not differentiable in d_1.

To scale egg production, Todd and Lintermans (2015) proposed to scale recruitment with a logistic function:

$$R_G(S, d_1) = R(S) \cdot \left(\frac{2}{1 + exp\left(-\frac{S}{d_1}\right)} - 1 \right).$$

This functional form is similar to the type D depensation above, but in this form, the width of the effect is fixed and the scaling is with S instead of $logS$. Note that the depensatory modification proposed by Todd and Lintermans (2015) used a sum of abundance-at-age instead of spawning biomass. Here, we present the model with spawning stock size to be consistent with the other models.

As a more specialised depensatory variation, Bal, Scheuerell, and Ward (2018) modified the hockey-stick recruitment model to include another breakpoint, thereby reducing the recruitment per spawner for low spawning stock biomass. Another recruitment model with multiple inflection points was used by Solari, MartinGonzalez, and Bas (1997) and Solari et al. (2010). They constructed a recruitment model from a number of equilibrium states in the stock–recruitment relationship such that:

$$R(S) = \sum_{i=1}^{n} \frac{a_i \cdot S}{\left(S - b_i\right)^2 + c_i},$$

where n is the number of equilibria. This recruitment model will have compensatory and depensatory phases between the equilibrium points. Finally, semiparametric models have been used to allow for depensatory recruitment, such as splines (Maroto and Moran 2014), GAM models (Langangen et al. 2011) and Gaussian processes (Sugeno and Munch 2013a, 2013b).

3.2.3 | Parameter Estimation and Uncertainty

The spawning stock biomass and recruitment pairs available as observations were model outputs from assessment models. As such, they are inherently uncertain and may contain quirky patterns influenced by model assumptions (e.g., Brooks and Deroba 2015; Dickey-Collas et al. 2014). Therefore, an errors-in-variables approach was used to estimate stock–recruitment curves. For each stock, s, and year, y, observed spawning stock biomass, $S_{s,y}$, was assumed to be log-normally distributed:

$$logS_{s,y} \sim N\left(log\widetilde{S}_{s,y}, \tau_{s,y}^2\right),$$

where $\widetilde{S}_{s,y}$ was the true, unobserved spawning stock biomass and $\tau_{s,y}$ was the standard error calculated from the assessment confidence interval. Likewise, observed recruitment, $R_{s,y}$, was assumed to be log-normally distributed:

$$logR_{s,y} \sim N\left(log\widetilde{R}_{s,y}, \upsilon_{s,y}^2\right),$$

where $\widetilde{R}_{s,y}$ was the true, unobserved recruitment and $\upsilon_{s,y}$ was the standard error calculated from the assessment confidence interval. Further, the true, unobserved recruitment was assumed to be related to the true, unobserved spawning stock biomass via the recruitment models described above:

$$log\widetilde{R}_{s,y} = logR\left(\widetilde{S}_{s,y}\right) + \epsilon_{s,y},$$

where $\epsilon_{s,y}$ were stock-wise stationary order one autoregressive processes with mean zero, autocorrelation parameter $-1 < \phi_s < 1$ and marginal variance σ_s^2. In the equation, $logR\left(\widetilde{S}_{s,y}\right)$ is the logarithm of the predicted recruitment based on the true, unobserved spawning stock biomass in year y. We refer to Data S5 for detailed equations on the stock–recruitment curves included. In this analysis, $\tau_{s,y}$ and $\upsilon_{s,y}$ were available from the assessment models, while ϕ_s and σ_s were estimated parameters. The true unobserved quantities, $\widetilde{S}_{s,y}$ and $\widetilde{R}_{s,y}$ were treated as latent variables.

Parameters were estimated using maximum likelihood. The marginal likelihood of observations were obtained through the Laplace approximation using the R package TMB (Kristensen et al. 2016, version 1.9.4). In the optimisation, parameters were penalised with an L_2 penalty (i.e., $\pi(p; a) = \frac{p^2}{a^2}$), where p is the parameter and a determines the strength of the penalisation. The parameters S_{max} and S_{50} were penalised on log scale towards the midpoint of the observed range (with $a = 5$), while R_{max} was penalised on log scale towards the highest observed value per stock (with $a = 5$). Remaining parameters were penalised towards zero on the scale of estimation (with $a = 0.5$). Parameter uncertainty was quantified using the observed Fisher information. Uncertainty of derived values was approximated using the delta method or through simulations (e.g., Aalen et al. 1997; Mandel 2013).

3.3 | Depensatory Thresholds

To describe the critical biomass for which depensation takes effect, three thresholds were calculated from the stock–recruitment curves. These thresholds were selected to highlight different aspects of the depensatory effects and may be used to define limit biomass reference points.

3.3.1 | Allee Effect Threshold

The first calculated threshold was the Allee effect threshold defined by Perala, Hutchings, and Kuparinen (2022). The threshold biomass is defined to be the inflection point where the stock–recruitment function changes from convex to concave. At this point, the second derivative is zero. Therefore, we calculate the biomass threshold, S_0 as

$$S_0 = argmin_S R''(S)^2.$$

For a compensatory recruitment model, the stock–recruitment function is concave for any spawning stock biomass. Consequently, $S_0 = 0$ for compensatory models.

3.3.2 | Maximum Productivity Threshold

The second calculated threshold was derived from the recruitment productivity of the stock. This threshold is defined as the biomass that maximises the reproductive rate of the stock:

$$S_{MP} = argmax_S \frac{R(S)}{S}.$$

Here, the reproductive rate is defined as the number of recruits produced per unit of spawning biomass. In general, other productivity measures could be applied. For instance, recruitment could be multiplied by the spawners produced per recruit, which would result in a different threshold when density dependence is present in maturity, mortality or growth. Similar to the Allee effect threshold, $S_{MP} = 0$ for compensatory recruitment models.

3.3.3 | Replacement Rate Threshold

The final calculated depensatory threshold, S_{NRR}, was defined as the inflection point where the replacement rate changes from negative to positive. The replacement rate was defined as:

$$r(S) = \frac{SPR(0) \cdot R(S) - S}{S},$$

where $SPR(0)$ is the equilibrium spawning biomass per recruit in the absence of fishing. For simplicity, $SPR(0)$ was calculated using biological data from the last available year. For a depensatory recruitment model, the replacement rate will be zero at both a low biomass, changing from negative to positive and at a high biomass, changing from positive to negative. Based on the definition, only the first point is of relevance. Therefore, we calculated the threshold as the point where the replacement rate was zero, but restrict the threshold to be between zero and the maximum productivity threshold. Combined, the replacement rate threshold can be derived as:

$$S_{NRR} = argmin_{0<S<S_{MP}} r(S)^2.$$

For compensatory recruitment models, the replacement rate is positive at low spawning biomass. Therefore, $S_{NRR} = 0$ for compensatory recruitment models. For all depensatory thresholds, estimation uncertainty was quantified using the delta method.

3.4 | Reference Points

For each model fit, deterministic reference points were derived from equilibrium values using per-recruit calculations assuming no stochasticity in recruitment and survival. The reference points included the equilibrium biomass in the absence of fishing, B_0, the maximum sustainable yield, MSY, the fishing mortality rate leading to the MSY, F_{MSY}, the equilibrium spawning stock biomass corresponding MSY, B_{MSY} and the smallest fishing mortality rate leading the stock to extinction, F_{ext}. For the compensatory models, F_{ext} corresponds to the smallest fishing mortality rate where the replacement line has no intercept with the recruitment curve, referred to as F_{Crash}. All reference points were based on biological data and fishing selectivity from the last available year. Calculations and criteria followed Albertsen and Trijoulet (2020). We also refer to Data S5 for details. Uncertainty of reference points was quantified using the delta method.

3.5 | Population Forecast

To evaluate the difference between expected stock rebuilding times in the absence of harvesting from depensatory and compensatory models, populations were forecasted using stochastic recruitment. For each stock, estimated recruitment models in combination with the biological data were used to forecast the population from different spawning stock biomass starting points corresponding to different levels of depletion (0.5%–90% of B_{MSY}). In the forecast, biological data from the last available year were used and no harvesting was assumed. That is, the fishing mortality rate was fixed to zero. First, starting SSB was converted to numbers-at-age using the same age composition as the last assessment year. In turn, the population was forecasted for 200 years using the stock-specific natural mortality and the fitted recruitment model. Stochastic recruitment was simulated from autocorrelated log-normal distributions where the median was determined by the stock–recruitment curve, using the estimated stock–recruitment, variance and autocorrelation parameters. We refer to Data S5 for detailed equations describing the stock–recruitment curves used. Since information about stochasticity in cohort survival was not available in the databases, cohort survival was projected with a deterministic exponential decay using the stock-specific natural mortality without fishing. To evaluate the effect of different depletion levels, each stock was forecasted with starting spawning stock biomasses $0.5, 1.0, 1.5, \ldots, 90\%$ of the compensatory deterministic B_{MSY}. For each stock and starting biomass value, the stochastic population forecast was replicated using 1500 iterations. An algorithmic description and forecast equations are given in Data S5.

3.6 | Forecast Summaries

Forecast simulations were summarised in terms of time to rebuild and probability of extinction under fishing moratorium.

For each simulation, time to rebuild was extracted as the minimum number of years for a simulated population to reach a spawning stock biomass of B_{MSY}, while the stock was considered extinct if spawning stock biomass came below one ton at any time. To summarise across simulations, the median rebuilding time and fraction of extinctions were calculated for each starting biomass. In turn, the biomass where rebuilding takes two generation times (e.g., Simpfendorfer 2005) to rebuild (denoted B^C_{2G} for the compensatory model and B^D_{2G} for the depensatory model), the biomass where the depensatory model takes half the generation time longer than the compensatory model to rebuild ($B_{+G/2}$) and the biomass with 5% probability of extinction were extracted ($B^C_{5\%}$ and $B^D_{5\%}$ respectively). Generation time was calculated as the mean period in years between birth of a parent and the birth of their offspring, following Simpfendorfer (2005), in the absence of fishing and using biological data from the last available year. To calculate confidence intervals and estimation uncertainty of forecast summaries that account for the parameter estimation uncertainty in the recruitment models, a simulation-based alternative to the delta method was used (e.g., Aalen et al. 1997; Mandel 2013).

3.7 | Meta-Analysis Across Life Histories

Depensatory thresholds, reference points, rebuilding time and probability of extinction were summarised across stocks in a meta-analysis. In the meta-analysis, average outcomes were calculated by fisheries guild and taxonomic family. Each of the guilds had a wide spatial distribution (Figure 1). Averages were calculated as inverse variance weighted averages with random effects (e.g., Hedges and Olkin 1985) on a suitable scale that accounted for the range of possible values. Reference points and thresholds were ensured to be positive, while rebuilding summaries were bounded between 0% and 100% of the compensatory B_{MSY}.

4 | Results

Recruitment models were fitted to the selected 79 Northeast Atlantic fish stocks and used for calculating depensatory thresholds, biological reference points and projecting rebuilding time. A total of 20 stock–recruitment models were fitted for each stock. In turn, the best-fitting stock–recruitment model, in terms of AIC, to available data was identified and used in the meta-analysis.

4.1 | Model Fits

Of the 79 stocks, 20 had the best fit from the Beverton–Holt family of models, 22 from the Ricker family, 6 from the Hassell/Deriso family and 31 from the hockey stick family of models. In most cases, the best-fitting model was the compensatory version. The type D depensatory model was the best fit for two herring (*Clupea harengus*, Clupeidae) stocks, while the type A depensatory model was the best fit for three demersal stocks; one combined with the hockey stick and two combined with the Ricker recruitment.

The power to detect depensatory recruitment depends on a combination of the effect size, the range of biomass observations, the number of stock–recruitment pairs and the uncertainty. Of the fitted stocks, 5.1% had all spawning stock biomass observations above 40% of the compensatory estimated B_0, while 16.5% had all observations above 20% of B_0, and 46.8% had all observations above 10% B_{MSY}. That is, most of the stocks have had periods where the stock would be considered overfished. For the compensatory fits, estimated B_{MSY} ranged from 1.0% to 57.2% of B_0 with a median of 28.4%. The first and third quantiles were at 21.1% and 37.7% respectively.

For the meta-analysis across stocks, the best-fitting model was retained. Within the best-fitting model family, the compensatory model was compared to the best-fitting depensatory model. For the majority of stocks, the best-fitting depensatory model, in terms of AIC, was the type D model, which was used for 58 stocks. The type A depensatory model was used for 17 stocks, and the type B was used for four stocks. The type C model was not used for any of the stocks.

4.2 | Loss of Recruitment

To quantify the difference in estimated recruitment between the depensatory and compensatory model fits, the ratio between them was calculated and summarised as a function of spawning biomass (Figure 4). At 10% of the compensatory B_{MSY}, the median reduction in estimated depensatory recruitment is 39.1% across all stocks. The reduction is at least 5% at biomass levels below 45.5% of B_{MSY} for benthic stocks, below 20.9% of B_{MSY} for demersal stocks and below 33.6% of B_{MSY} for pelagic stocks.

4.3 | Reference Points

Estimated deterministic reference points were compared between the compensatory and the best depensatory recruitment model fits from the best-fitting recruitment model family in a meta-analysis (Figure 5). Across all life histories, F_{MSY} was found to be 9.0% (95% confidence interval (CI): 5.8%, 12.2%) larger for the depensatory models than for the compensatory models. The difference was larger for pelagic stocks (22.9% on average; CI: 18.6%, 27.4%). In contrast, B_{MSY} was lower for depensatory models than for compensatory models (−5.6% on average; CI: −10.9%, −0.1%). Consequently, maximum sustainable yield was slightly larger for depensatory models when averaging over life histories (3.0% on average; CI: 0.0%, 6.1%). Biologically, F_{MSY} is not expected to increase in the presence of depensation, all other parameters being unchanged. Depensation affects the yield curve at high fishing mortality rates, where equilibrium spawning biomass is low. The lower yield at high fishing mortality rates will shift F_{MSY} to a lower value. However, when deriving recruitment dynamics and reference points from data, all other parameters are not unchanged. The depensation parameters are estimated together with other stock–recruitment relation parameters. In the presence of depensation, a compensatory recruitment model will overestimate the reproductive rate at low biomass by construction. Since the estimation procedure aims to give a good fit on average, the model often compensates by

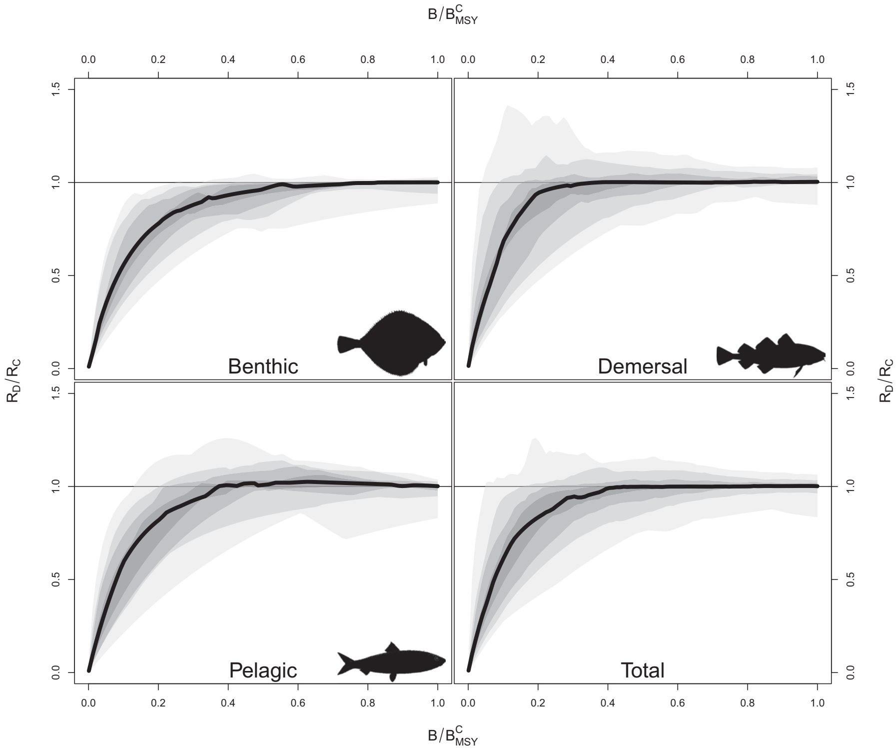

FIGURE 4 | Quantiles of the relative difference in predicted recruitment between the compensatory (R_C) and depensatory (R_D) model fits as a function of spawning stock biomass (B) relative to the compensatory B_{MSY}. Values below one indicate that the depensatory model estimates lower recruitment. The lightest grey region indicates the range between the 10% and 90% quantile, the second lightest grey region indicates the range between the 20% and 80% quantiles and so on. The full black line indicates the median across stocks.

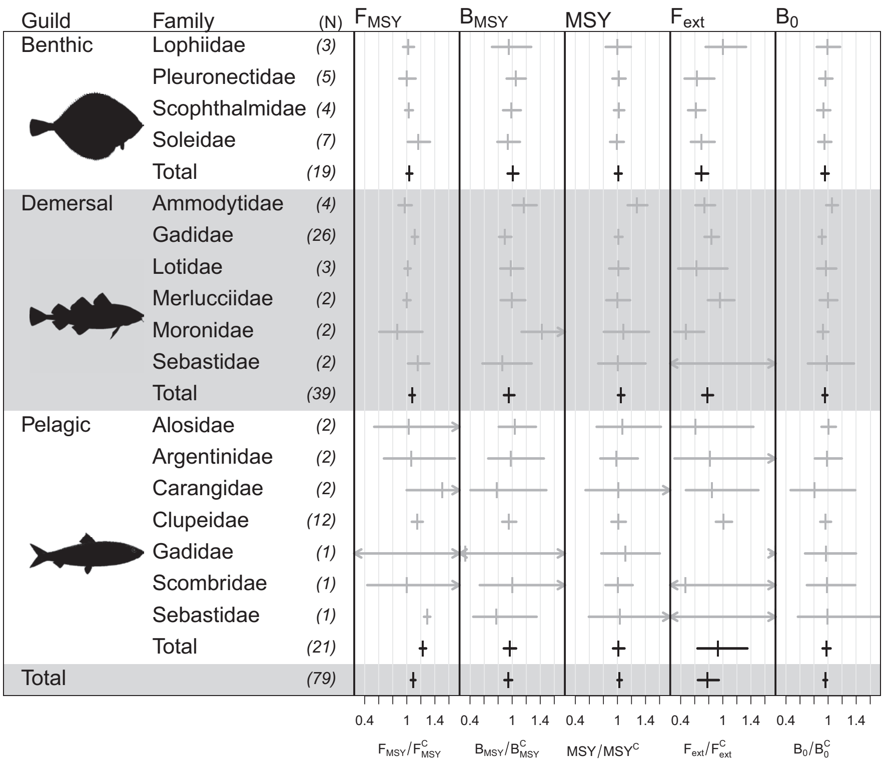

FIGURE 5 | Inverse variance weighted average fraction of depensatory F_{MSY}, B_{MSY}, MSY, F_{ext} and B_0 to the corresponding compensatory estimate by fisheries guild and family. Estimates are indicated by a vertical line while 95% confidence intervals are indicated by a horizontal line. N is the number of stocks.

underestimating the reproductive rate at medium biomass. When a depensation parameter is included, the model can get a closer fit to data at medium biomass, which may lead to higher yield. As a result, the estimated F_{MSY} may increase compared to the compensatory fit.

For all guilds, including depensation in the recruitment model resulted in a lower fishing mortality leading to extinction, F_{ext}. Averaging across benthic life histories, F_{ext} was reduced by 30.1% (CI: 20.9%, 38.3%), while the reduction was 21.9% (CI: 13.9%, 29.2%) for demersal stocks. For pelagic life histories, the average decrease of 7.1% (CI: −35.0%, 36.0%) was not statistically significant. For the largest pelagic family, the clupeids, there was essentially no difference in F_{ext} with an average increase of 1.0% (CI: −9.8%, 13.1%). Further, a small decrease of 4.1% (CI: 1.5%, 6.6%) in the estimated unfished equilibrium biomass was found for the depensatory models.

Looking only at the five stocks where a depensatory model had the lowest AIC, similar results were obtained. In this case, F_{MSY} was 9.3% (CI: −0.1%, 19.6%) higher for the depensatory models than for the compensatory models, B_{MSY} was decreased by 0.7% (CI: −6.8%, 7.7%), resulting in a 10.0% (CI: 1.2%, 19.6%) increase in MSY. Further, a decrease of 15.5% (CI: 5.2%, 24.7%) was seen for F_{ext} and of 6.5% (CI: 0.9%, 11.8%) for B_0.

The overall results were largely unaffected by recruitment families (Data S1–S4). Assuming the same recruitment model for all stocks resulted in similar conclusions for F_{MSY}, B_{MSY}, MSY and B_0. For F_{ext}, a difference was observed for the hockey stick recruitment model. Assuming a hockey stick model for all stocks resulted in a 40% increase in F_{ext} for the depensatory fits across all stocks (Data S4). The increase was mainly driven by the pelagic stocks, while the increase was small for demersal stocks. Benthic stocks had similar F_{ext} for the compensatory and depensatory models. A similar conclusion was obtained when analysing only stocks where the hockey stick was the best-fitting model.

In the hockey stick model, the linear increase in recruitment with spawning biomass below the breakpoint results in a sharp drop in equilibrium biomass and yield as a function of fishing mortality rate. For some stocks, the drop can happen with fishing mortality rates below the rate leading to the maximum yield per recruit. In this case, $F_{\mathrm{MSY}} = F_{\mathrm{ext}} = F_{\mathrm{crash}}$. For the stocks where the hockey stick provided the best fit, the compensatory F_{MSY} ranged from 12.3% to 85.3% of F_{ext} with a median of 53.7%. For the three other recruitment families, F_{MSY} ranged from 2.0% to 69.4% with a median of 34.7% of F_{ext}. For most stocks, the relative difference between F_{MSY} and F_{ext} was smaller for the depensatory than the compensatory fits. This was, however, not the case for 0.4% of stocks where the hockey stick provided the best fit to data.

4.4 | Depensatory Thresholds

For each stock, three depensatory thresholds were calculated (Figure 6). Following Perala, Hutchings, and Kuparinen (2022), the Allee effect threshold (S_0), where the recruitment model changes from concave to convex, was calculated. Averaging across all life histories, the threshold was found to be at 21.6% (CI: 19.7%, 23.7%) of the compensatory B_{MSY}. The threshold was higher for pelagic fish at 27.4% (CI: 23.0%, 32.7%) than for demersal at 18.9% (CI: 16.5%, 21.6%) and benthic stocks at 20.5% (CI: 16.5%, 25.5%).

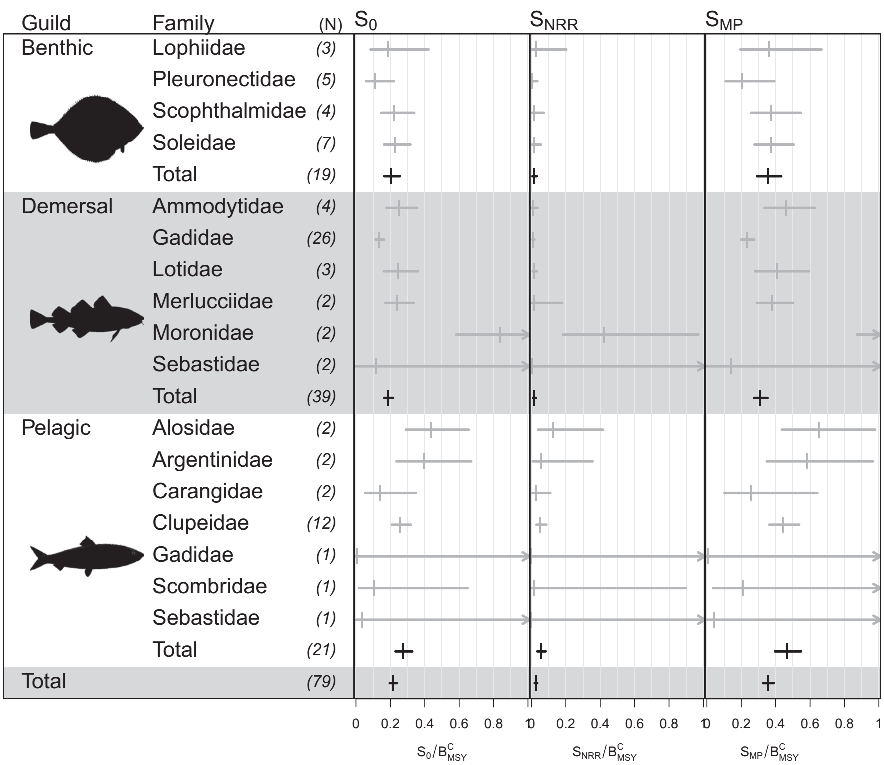

FIGURE 6 | Inverse variance weighted average fraction of depensatory thresholds S_0, S_{NRR} and S_{MP} to the compensatory B_{MSY} by fisheries guild and family. Estimates are indicated by a vertical line while 95% confidence intervals are indicated by a horizontal line. N is the number of stocks.

Besides the Allee effect S_0 threshold, the spawning stock biomass with maximum reproductive rate, S_{MP}, was estimated. Across all life histories, the point of the maximum reproductive rate was found at 35.7% (CI: 32.6%, 38.9%) of the compensatory B_{MSY}. The threshold was slightly lower for benthic stocks at 35.5% (CI: 29.1%, 43.2%) and demersal stocks at 31.1% (CI: 27.5%, 35.2%), while it was found to be at 46.6% (CI: 39.6%, 54.7%) of the compensatory B_{MSY} for the pelagics. Further, the threshold at the biomass where the replacement rate changes from negative to positive (S_{NRR}) was estimated to be at very low biomass compared to B_{MSY} when averaging over all life histories (2.5% on average; CI: 1.9%, 3.4%).

Considering only the five stocks where a depensatory model had the lowest AIC, thresholds were estimated to be higher. For this subset of stocks, the Allee effect threshold, S_0, was at 44.4% (CI: 39.1%, 50.3%) of B_{MSY} on average, S_{MP} was at 64.6% (CI: 57.6%, 72.3%), while S_{NRR} was at 8.1% (CI: 4.8%, 13.5%).

The depensatory thresholds were affected by the recruitment model. If Ricker recruitment was assumed for all stocks, average thresholds were similar to those obtained with the best-fitting recruitment model, while assuming Beverton–Holt or Hassell/Deriso models for all stocks tended to have lower thresholds, and the hockey stick model had higher thresholds (Data S1–S4).

Of the 19 benthic stocks, five have been below S_{MP} at some point, while three have been below S_0. For pelagic stocks, 11 of 21 stocks have been below S_{MP}, and seven have been below S_0, while 24 and 11 of the 39 demersal stocks have been below S_{MP}, respectively, S_0. In the last data year, one benthic, seven demersal and four pelagic stocks are below S_{MP}, while four demersal and two pelagic stocks are below S_0.

4.5 | Rebuilding From Low Biomass

The last part of the meta-analysis evaluated the difference in rebuilding potential between compensatory and depensatory recruitment (Figure 7). For the compensatory model fits, 68 of the 79 models forecast showed no risk of stochastic extinction for any starting biomass level, one haddock and three sandeel stocks had more than 5% risk for all starting biomass levels, while one cod stock had more than 5% risk of extinction below 2% of B_{MSY}. In contrast, 17 depensatory recruitment fits showed no risk of extinction, while nine of the stocks had more than 5% risk of extinction for all starting biomass levels up to 90% of B_{MSY}. On average, benthic populations had the lowest risk of extinction. For these, the biomass with 5% risk was found at 1.2% (CI: 0.3%, 4.5%) of the compensatory B_{MSY}. The biomass with 5% risk of extinction was higher for the demersal stocks (25.8% on average; CI: 22.6%, 29.3%), mostly driven by the sandeels (89.6% on average; CI: 87.4%, 91.5%), followed by the pelagics which had the highest average biomass with 5% risk of extinction (41.8% on average; CI: 31.2%, 53.1%). These results indicate that an average pelagic stock with assumed compensatory recruitment would be perceived to be able to recover with no risk of depletion if fishing is stopped. However, if the true system is depensatory, there would 5% risk of collapse in the absence

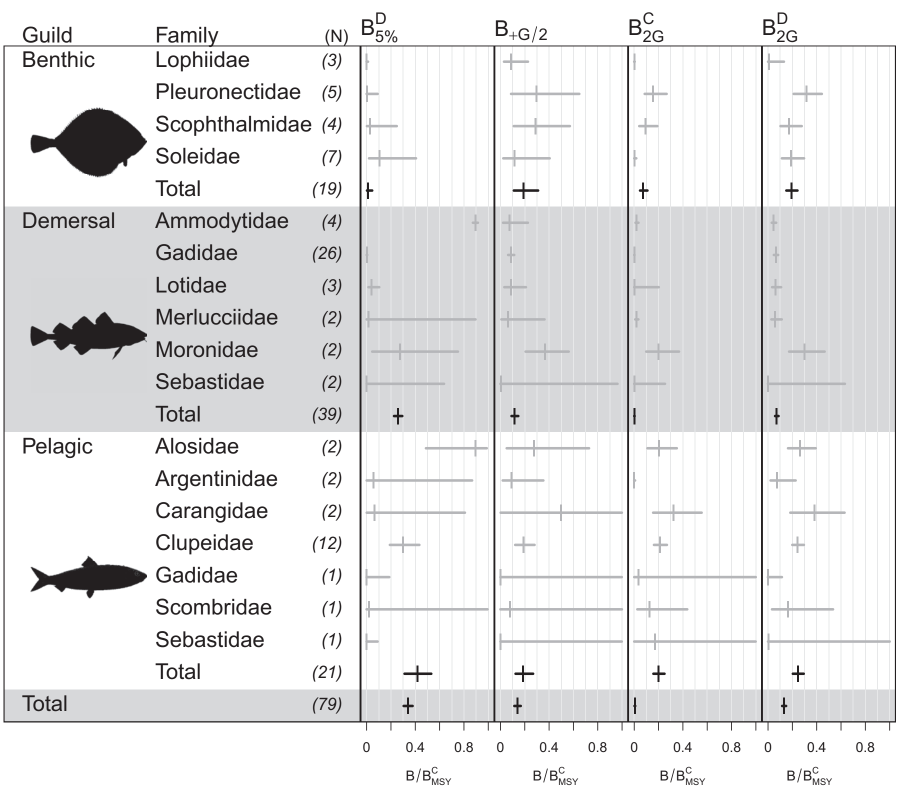

FIGURE 7 | Inverse variance weighted average fraction of rebuilding summaries $B^D_{5\%}$, $B_{+G/2}$, B^C_{2G}, B^D_{2G} to the corresponding compensatory estimate by fisheries guild and family. Estimates are indicated by a vertical line while 95% confidence intervals are indicated by a horizontal line. N is the number of stocks.

of fishing at 40% of the compensatory estimated B_{MSY}. Under depensation, the risk increases as the stock biomass decreases, while the compensatory perception continues to be that there is no risk of depletion.

Besides increasing the risk of stochastic extinction, the depensatory model increased rebuilding times from low biomass. In terms of rebuilding time, the two perceptions of productivity were compared on the biomass where rebuilding to B_{MSY} took two generations, as well as the point where the depensatory model took one generation longer to rebuild than the compensatory model. Pelagic populations had the highest average biomass with two generations to rebuild for both compensatory and depensatory models, but also the smallest difference. The compensatory models took two generations to rebuild from biomasses below 19.9% (CI: 15.8%, 24.9%) of the compensatory B_{MSY} while it was below 24.6% (CI: 20.4%, 29.3%) of the compensatory B_{MSY} for depensatory models. On average, benthic stocks took two generations to rebuild from 7.3% (CI: 4.8%, 10.8%) of the compensatory B_{MSY} for the compensatory perception of recruitment, which increased to 19.2% (CI: 15.1%, 24.2%) of B_{MSY} for the depensatory models. The relatively fastest rebuilding compared to generation time was found for demersal populations. With the compensatory recruitment assumption, rebuilding took two generations from 0.1% (CI: 0.1%, 0.3%) of the compensatory B_{MSY} on average, which increased to 7.1% (CI: 5.8%, 8.6%) for the depensatory models.

Averaging over all life histories, depensatory recruitment increased rebuilding time by at least half a generation at biomass below 13.9% (CI: 11.5%, 16.7%) of the compensatory B_{MSY}. The averages were similar for all three guilds.

When only looking at stocks where the depensatory fit had the lowest AIC, the starting biomass fraction of B_{MSY} with 5% risk of stochastic extinction was lower (20.0% on average; CI: 11.1%, 33.3%), while the level with two generation times to rebuild for the depensatory model (12.3% on average; CI: 9.1%, 16.6%) as well as the level where the depensatory model took half a generation longer to rebuild (16.4% on average; CI: 14.1%, 18.9%) were similar to the results for all stocks.

All rebuilding diagnostics were affected by recruitment models. The biomass with 5% risk of extinction for depensatory fits were lower when assuming Beverton–Holt recruitment, and close to zero for benthic and demersal species. Assuming either Hassell/Deriso or hockey stick recruitment also resulted in a lower 5% risk threshold across life histories, driven by demersal fish, while the risk threshold increased for the benthic species. Assuming Ricker recruitment gave a larger 5% risk of extinction threshold driven by a large increase for benthic and demersal stocks. Across guilds, the biomass where rebuilding time increased by half a generation was larger when assuming Beverton–Holt (except for pelagic species), Ricker or Hassell/Deriso recruitment. Assuming hockey stick recruitment resulted in lower values for all three guilds. Overall, the biomass where rebuilding took two generations was largely unaffected by the recruitment models (See Data S1–S4). The only exception was that rebuilding time increased for benthic stocks with Beverton–Holt depensatory recruitment (see Data S1).

5 | Discussion

Two central aspects of fisheries stock assessment and scientific advice require evaluating (i) sustainable fishing levels and (ii) options for recovering stocks from a poor biological state (i.e., low level of stock spawning biomass). Both objectives depend on the knowledge of the stock–recruitment relationship. Currently, most stock–recruitment relationships used in management are assumed to have the compensatory mortality property where the recruitment per spawner rate is assumed to grow with decreasing stock size (Albertsen and Trijoulet 2020; ICES 2021a; Kolody et al. 2019; e.g., Methot and Wetzel 2013; Nielsen and Berg 2014; Sharma et al. 2019). In contrast, models with depensation or Allee effects will have decreasing recruitment rates below a certain stock size.

5.1 | Impact of Depensatory Dynamics on Reference Points and Rebuilding

Through a meta-analysis of 79 Northeast Atlantic fish stocks, covering several ecosystems and taxonomic families, the effect of assuming a compensatory recruitment relationship if the Allee effect is present was investigated. By construction, compensatory recruitment models have increasing productivity as the stock size declines. In contrast, this meta-analysis found that the maximum reproductive rate is at 36% of the compensatory B_{MSY} on average across life histories with the depensatory recruitment assumption. For pelagic populations, the maximum reproductive rate was 47% of B_{MSY} on average. Recently, Perala, Hutchings, and Kuparinen (2022) suggested the level of spawning biomass where the recruitment curve transitions from convex to concave as the Allee effect threshold. Averaging across life histories, the Allee effect threshold was found at 22% of B_{MSY}. The threshold was slightly higher for pelagic stocks, and slightly lower for benthic and demersal stocks. One should be particularly careful when a stock is approaching these biomass levels as it could lead to nonprecautionary management if potential depensatory effects are not considered. Therefore, those SSB levels estimated for each guild might be considered as a starting point for setting minimum thresholds of limit reference points.

For target reference points, the recruitment assumption had limited influence. In particular, estimated MSY and equilibrium unfished biomass were largely unaffected. However, the constant fishing mortality rate leading to MSY, F_{MSY}, was estimated to be 9% larger for depensatory models, whereas the corresponding equilibrium biomass, B_{MSY}, was 5.6% lower. In theoretical studies, the opposite is often found (e.g., Marks et al. 2021) if other population parameters remain unchanged. However, when estimating population dynamics from data, all parameters will be updated simultaneously, which can lead to a lower estimated recruitment productivity at low biomass, a higher productivity at medium biomass and similar productivity at high biomass when including depensation in the recruitment model. Further, the lowest fishing mortality rate leading to equilibrium extinction was reduced by 22% on average. Combined, this suggests that under depensation, stocks should be fished harder to obtain the maximum yield, but there is a higher vulnerability to recruitment over-fishing. As a result, a precautionary management could use the lower of the compensatory and depensatory estimated target reference points while ensuring that harvest control rules limit the risk of reducing stock size to the critical levels estimated by assuming depensatory dynamics. The latter would require an earlier and stronger response to reductions in stock biomass than what would be inferred from compensatory recruitment models.

Rebuilding stocks from depleted biomass levels was also affected by the presence of the Allee effect. For pelagic stocks, rebuilding was found to take at least two generations below 25% of B_{MSY} on average for depensatory models, while it was at 20% for the compensatory models. Likewise, this biomass point increased from 7% to 19% of B_{MSY} for benthic stocks, and from $< 1\%$ to 7% for demersal stocks. Across life histories, rebuilding time was found to increase by half a generation time below 14% of the compensatory B_{MSY}.

These effects of depensation are seemingly in contrast with earlier studies such as Hilborn et al. (2014), which concluded that depensation is not a concern for rebuilding or limiting reference points. However, the conclusion of, for example, Hilborn et al. (2014) is based on the premise that if depensation is not statistically significant, the compensatory alternative is a better description of the rebuilding potential. On the other hand, the purpose of this study was to investigate the consequence of assuming compensatory recruitment if depensation is present, even if not statistically significant, thus conducting robustness test on the effect of a different structural assumption of the shape of the stock and recruitment relationship at low level of spawning biomass.

5.2 | Detecting Depensatory Dynamics

This study explored the consequences of depensatory dynamics in fisheries management by relying on available data to inform the stock–recruitment models. However, it is essential to acknowledge the challenges of detecting depensatory dynamics in real-world fisheries data (Hilborn et al. 2014; Liermann and Hilborn 1997; e.g., Myers et al. 1995). In this study, only 26 of the 79 stocks have observations at low stock biomass, that is, below 20% of B_{MSY}. In practice, data limitations often lead to a situation where the analysis leans towards compensatory dynamics because the presence of depensatory dynamics may not be evident from the available data, due to a lack of observations at low abundances. This is particularly important when the fit is evaluated using measures such as AIC. While AIC can be useful for determining the structure of assessment models (e.g., Albertsen, Nielsen, and Thygesen 2017, 2018), including recruitment (Trijoulet et al. 2022), it will be limited to evaluate the fit within the available range of data. Therefore, several observations at low abundance may be necessary for AIC to select a depensatory recruitment curve over a compensatory model with fewer parameters. In spite of these limitations, a depensatory model was the best fit for five stocks, three of which did not have observations below 20% of B_{MSY}. Considering only these five stocks, and accounting for their fisheries guilds, conclusions were generally similar to the full meta-analysis. However, the stocks where the depensatory model was the best fit tended to have higher depensatory thresholds, indicating a need to act even earlier on overfishing. Besides the available data range, the inability to detect significant depensation can be caused by several factors, including environmental changes and data limitations.

Our study did not account for environmental effects such as variations in temperature, food availability and predation pressure, which have all been identified as key drivers of environment-driven recruitment variability (e.g., Brunel and Boucher 2007; Fauchald 2010; Okamoto et al. 2012; Olsen et al. 2010). Moreover, we also make the implicit assumption that future recruitment can be predicted by observations made in the past, but it has been recognised that marine ecosystems can experience 'regime shifts' (deYoung et al. 2008; Vasilakopoulos and Marshall 2015), which may result in changes in fish population dynamics (Beaugrand 2004), and translate into rapid changes or nonstationarity in the parameters of the stock–recruitment relationships (Perälä, Swain, and Kuparinen 2017). On the one hand, such changes in recruitment may mask depensatory effects (Tirronen, Perala, and Kuparinen 2022). On the other hand, environmental conditions may influence recruitment to a degree that drives changes in stock abundance (e.g., Szuwalski et al. 2014). Such changes in recruitment may result in regime shifts that can create spurious correlations between spawning stock size and recruitment and make stock–recruitment curves appear depensatory. Since this meta-analysis did not consider environmental effects, fitted recruitment curves for individual stocks reflect the average productivity over environmental and productivity regimes experienced in the past. While appropriate for synthesis across life histories and ecosystems, assessment of individual stocks should account for potential productivity changes when modelling recruitment.

The data used in this meta-analysis are the output of fish stock assessments. These assessments are largely aligned with management purposes. Therefore, they may be misaligned with spawning populations, which can influence the ability to detect depensation. Some of the stocks in this study are recognised to be a mixture of individual populations in all or part of the assessment areas, such as the herring stocks (Bekkevold et al. 2023; e.g., Farrell et al. 2022), North Sea cod (e.g., ICES 2022b, 2023a) and Baltic cod stocks (Albertsen et al. 2021; e.g., Hüssy et al. 2016, 2022). If depensatory recruitment exists for substocks, the effect can be masked when assessing stocks as a complex (Frank and Brickman 2000). This could potentially lead to nonprecautionary management of the individual populations.

5.3 | Limitations of the Available Data

Even for stocks where the assessment and spawning dynamics are aligned, the stock–recruitment data used are based on stock assessment model outputs and may reflect model assumptions, including fitted stock–recruitment relationships (Dickey-Collas et al. 2014). If not accounted for, this aspect introduces potential biases or uncertainties into subsequent analyses (Brooks and Deroba 2015). To mitigate the effects of fitting recruitment curves to assessment model output, an errors-in-variables approach was used to account for the model uncertainty in 'observed' spawning biomass and recruitment. The approach incorporated the uncertainty of stock–recruitment pairs to give each pair the correct weight when fitting the model. As a result, the uncertainty of estimated recruitment curve parameters is more accurate and, in turn, so is the uncertainty in forecast and results.

While the errors-in-variables approach improves model fitting, the analysis was limited by the available model estimates of uncertainty and the lack of correlation estimates between spawning biomass and recruitment as well as temporal correlation. We recognise that stock assessment model estimates are correlated to some degree. However, lack of information necessitated an assumption of independence over time in spawning biomass and between observed spawning biomass and recruitment. In our analysis, temporal correlation in recruitment was modelled. The assumption of independence can lead to underestimation of uncertainty in the model parameters and overconfidence in the subsequent analysis of rebuilding times and probabilities of extinction. Further, including too little uncertainty in estimated stock–recruitment pairs in the form of an over-precise errors-in-variables approach could favour stock–recruitment curves that mimic the curve assumed in the assessment. The stock–recruitment relationship assumed in the assessment was not available in any of the databases used. However, in our study, recruitment curves that are not implemented in the assessment model used were selected for some stocks. Likewise, assessments using the SAM model often apply a random-walk recruitment model, which we did not include in this study. Therefore, we do not expect this potential issue to influence the present meta-analysis.

Similar to stock abundance and recruitment, biological input data are model output. Since information on uncertainty was not available, the population forecasts assumed that the biological input data, including vital parameters and life history traits, were known (i.e., perfect knowledge). However, the reality of fisheries management often presents complexities and uncertainties that challenge this assumption. Stock boundaries, for instance, can be ambiguous, leading to potential misclassifications and uncertainties in stock assessments (Berger et al. 2020). Furthermore, the accuracy of biological parameters, such as the average weight of individuals, natural mortality rates and maturity, can vary not only between species but also among different populations within a species. Errors or inaccuracies in these crucial parameters can propagate throughout the analysis, potentially resulting in misleading predictions and management recommendations (García-Carreras, Jennings, and Le Quesne 2015). Robust methods for quantifying and reducing uncertainties associated with stock definitions and biological parameters should be a priority in future research.

Another important consideration is the inherent variability in the age of recruitment and productivity levels among different fish stocks. The parameters of recruitment functions are not directly comparable between stocks due to differences in life-history traits, environmental conditions and fishing pressures. Our analysis strives to provide a general understanding of depensation and its potential consequences across a diverse set of fish stocks. Still, it is crucial to recognise that the intricacies of recruitment patterns and productivity levels can vary significantly from one stock to another. To address this variability, future studies should explore ways to account for stock-specific differences in recruitment dynamics and productivity. Tailoring management strategies to the specific characteristics of each stock could lead to more effective and sustainable fisheries management practices. Additionally, the collection of detailed, stock-specific data on recruitment patterns and life-history traits will be vital for improving our understanding of these variations and their implications for depensation.

5.4 | Incorporating Depensatory Recruitment in Assessment and Advice

The limitations to detecting depensation from real-world fisheries data is critical for understanding the true implications of depensation in fisheries in terms of reference points, target reference points, rebuilding times and the probability of stock extinction. To better address these challenges, future studies, as well as scenarios used in assessments and advice, could utilise simulations assuming various strengths for the Allee effect. Simulation studies provide a controlled environment to explore the consequences of depensatory dynamics when data limitations hinder their detection. Further, several approaches to incorporating depensation should be investigated. Due to the complexity of recruitment processes in marine environments, deriving depensation models from first principles may not provide the best fit to data. In this study, the two-parameter model (Type D) generally performed better than other previously published models. In contrast, the model derived from the probability of mating (Type C, Hilborn et al. 2014) was not selected as the best-fitting depensatory model for any of the stocks.

In this study, the focus has been on the difference in perceptions of the stocks when assuming compensatory and depensatory recruitment. While the depensatory models were rarely selected by AIC, this may be caused by low statistical power (e.g., Sugeno and Munch 2013b). Given the potential consequences of misclassifying depensatory recruitment as compensatory highlighted in this meta-analysis, caution should be exercised before dismissing the possibility of depensation. Besides stock assessment, optimised harvest control rules are central to fisheries management. The impact of depensation on harvesting strategies was outside the scope of this study. However, previous theoretical studies have constructed optimised harvest control rules for certain depensatory population dynamics (Chan and Kim 2014; Çifdalöz 2022; Flores et al. 2007; Golden, van Poorten, and Jensen 2022; e.g., Gonzalez-Duran et al. 2018; Shertzer and Prager 2007; Vasconcellos 2003). In theoretical studies, the impact of interactions between parameters when estimated from data is seldomly considered. In this meta-analysis, we found that F_{MSY} increased on average when including depensation in recruitment, while B_{MSY} decreased. This would indicate that fish stocks can be fished harder when assuming depensatory dynamics, but this is associated with a higher risk of collapse. Therefore, harvest control rules to optimise yield while minimising risk could be evaluated under both depensatory and compensatory stock recruitment assumptions. Nonetheless, further studies on the impact on harvesting strategies are needed.

Through a meta-analysis of 79 Northeast Atlantic fish stock, this study is a synthesis of the effects of depensation on rebuilding and reference points across life histories and ecosystems. The results suggest that, on average, assuming compensatory recruitment dynamics may provide over-optimistic rebuilding projections and risk assessments at biomass below 25% of B_{MSY} in general, and below 45% for pelagic stocks. Notably, conclusions were robust whether considering all stocks or only stocks where a depensatory model was the best fit in terms of AIC. As a result, the possibility of depensation, even if not statistically significant, should be considered for precautionary fisheries assessment, advice and management. When incorporating depensation in recruitment models for individual stocks, the model should be tailored to the specific stock and thoroughly validated using residuals (e.g., Thygesen et al. 2017; Trijoulet et al. 2023) and goodness-of-fit (e.g., Albertsen, Nielsen, and Thygesen 2017). However, even if a compensatory model provides a better fit to the available data, potential effects of depensation should be considered in forecast scenarios of future fishing opportunities, risk evaluations and rebuilding strategies.

Acknowledgements

This work was funded by the European Maritime and Fisheries Foundation and the Ministry of Environment and Food of Denmark for the project 'Rebuilding Western Baltic Spring spawning herring' (grant ID: 33113-B-20-174), and the Academy of Finland (project grant 340901), In addition, this project has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation programme (grant agreement No 770884). The present study reflects only the authors' view, and the European Research Council is not responsible for any use that may be made of the information it contains.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

All data used are available from public sources referenced in the manuscript.

References

Aalen, O. O., V. T. Farewell, D. de Angelis, N. E. Day, and O. Nöel Gill. 1997. "A Markov Model for HIV Disease Progression Including the Effect of HIV Diagnosis and Treatment: Application to AIDS Prediction in England and Wales." *Statistics in Medicine* 16, no. 19: 2191–2210. https://doi.org/10.1002/(SICI)1097-0258(19971015)16:19<2191::AID-SIM645>3.0.CO;2-5.

Albertsen, C. M., K. Hüssy, S. H. Serre, J. Hemmer-Hansen, and T. B. Thomsen. 2021. "Estimating Migration Patterns of Fish From Otolith Chemical Composition Time-Series." *Canadian Journal of Fisheries and Aquatic Sciences* 78, no. 10: 1512–1523. https://doi.org/10.1139/cjfas-2020-0356.

Albertsen, C. M., A. Nielsen, and U. H. Thygesen. 2017. "Choosing the Observational Likelihood in State-Space Stock Assessment Models." *Canadian Journal of Fisheries and Aquatic Sciences* 74, no. 5: 779–789. https://doi.org/10.1139/cjfas-2015-0532.

Albertsen, C. M., A. Nielsen, and U. H. Thygesen. 2018. "Connecting Single-Stock Assessment Models Through Correlated Survival." *ICES Journal of Marine Science* 75, no. 1: 235–244. https://doi.org/10.1093/icesjms/fsx114.

Albertsen, C. M., and V. Trijoulet. 2020. "Model-Based Estimates of Reference Points in an Age-Based State-Space Stock Assessment Model." *Fisheries Research* 230: 105618. https://doi.org/10.1016/j.fishres.2020.105618.

Bal, G., M. D. Scheuerell, and E. J. Ward. 2018. "Characterizing the Strength of Density Dependence in At-Risk Species Through Bayesian Model Averaging." *Ecological Modelling* 381: 1–9. https://doi.org/10.1016/j.ecolmodel.2018.04.012.

Barrowman, N. J., and R. A. Myers. 2000. "Still More Spawner-Recruitment Curves: The Hockey Stick and Its Generalizations." *Canadian Journal of Fisheries and Aquatic Sciences* 57, no. 4: 665–676. https://doi.org/10.1139/f99-282.

Barrowman, N. J., R. A. Myers, R. Hilborn, D. G. Kehler, and C. A. Field. 2003. "The Variability Among Populations of Coho Salmon in the Maximum Reproductive Rate and Depensation." *Ecological Applications* 13, no. 3: 784–793.

Beaugrand, G. 2004. "The North Sea Regime Shift: Evidence, Causes, Mechanisms and Consequences." *Progress in Oceanography* 60, no. 2–4: 245–262. https://doi.org/10.1016/j.pocean.2004.02.018.

Bechtol, W. R., and G. H. Kruse. 2009. "Analysis of a Stock-Recruit Relationship for Red King Crab off Kodiak Island, Alaska." *Marine and Coastal Fisheries* 1, no. 1: 29–44. https://doi.org/10.1577/C08-052.1.

Bekkevold, D., F. Berg, P. Polte, et al. 2023. "Mixed-Stock Analysis of Atlantic Herring (*Clupea harengus*): A Tool for Identifying Management Units and Complex Migration Dynamics." *ICES Journal of Marine Science* 80, no. 1: 173–184. https://doi.org/10.1093/icesjms/fsac223.

Benoît, H. P., D. P. Swain, W. D. Bowen, G. A. Breed, M. O. Hammill, and V. Harvey. 2011. "Evaluating the Potential for Grey Seal Predation to Explain Elevated Natural Mortality in Three Fish Species in the Southern Gulf of St. Lawrence." *Marine Ecology Progress Series* 442: 149–167.

Berger, A. M., J. J. Deroba, K. M. Bosley, et al. 2020. "Incoherent Dimensionality in Fisheries Management: Consequences of Misaligned Stock Assessment and Population Boundaries." *ICES Journal of Marine Science* 78, no. 1: 155–171. https://doi.org/10.1093/icesjms/fsaa203.

Boettiger, C., D. Temple Lang, and P. Wainwright. 2012. "Rfishbase: Exploring, Manipulating and Visualizing FishBase Data From r." *Journal of Fish Biology* 81: 2030–2039. https://doi.org/10.1111/j.1095-8649.2012.03464.x.

Brooks, E. N., and J. J. Deroba. 2015. "When "Data" Are Not Data: The Pitfalls of Post Hoc Analyses That Use Stock Assessment Model Output." *Canadian Journal of Fisheries and Aquatic Sciences* 72, no. 4: 634–641. https://doi.org/10.1139/cjfas-2014-0231.

Brunel, T., and J. Boucher. 2007. "Long-Term Trends in Fish Recruitment in the North-East Atlantic Related to Climate Change." *Fisheries Oceanography* 16, no. 4: 336–349. https://doi.org/10.1111/j.1365-2419.2007.00435.x.

Cabral, R. B., P. M. Alino, and M. T. Lim. 2013. "A Coupled Stock-Recruitment-Age-Structured Model of the North Sea Cod Under the Influence of Depensation." *Ecological Modelling* 253: 1–8. https://doi.org/10.1016/j.ecolmodel.2012.12.031.

Chan, M. H., and P. S. Kim. 2014. "Modelling the Impact of Marine Reserves on a Population With Depensatory Dynamics." *Bulletin of Mathematical Biology* 76, no. 9: 2122–2143. https://doi.org/10.1007/s11538-014-0003-2.

Chen, D., J. Irvine, and A. Cass. 2002. "Incorporating Allee Effects in Fish Stock-Recruitment Models and Applications for Determining Reference Points." *Canadian Journal of Fisheries and Aquatic Sciences* 59, no. 2: 242–249. https://doi.org/10.1139/F02-005.

Çifdalöz, O. 2022. "Sustainable Management of a Renewable Fishery Resource With Depensation Dynamics From a Control Systems Perspective." *Gazi University Journal of Science* 35, no. 3: 936–955.

Courchamp, F., L. Berec, and J. Gascoigne. 2008. *Allee Effects in Ecology and Conservation*. Oxford: Oxford University Press. https://doi.org/10.1093/acprof:oso/9780198570301.001.0001.

Deriso, R. B. 1980. "Harvesting Strategies and Parameter Estimation for an Age-Structured Model." *Canadian Journal of Fisheries and Aquatic Sciences* 37, no. 2: 268–282. https://doi.org/10.1139/f80-034.

deYoung, B., M. Barange, G. Beaugrand, et al. 2008. "Regime Shifts in Marine Ecosystems: Detection, Prediction and Management." *Trends in Ecology & Evolution* 23, no. 7: 402–409. https://doi.org/10.1016/j.tree.2008.03.008.

Dickey-Collas, M., N. T. Hintzen, R. D. M. Nash, P.-J. Schön, and M. R. Payne. 2014. "Quirky Patterns in Time-Series of Estimates of Recruitment Could Be Artefacts." *ICES Journal of Marine Science* 72, no. 1: 111–116. https://doi.org/10.1093/icesjms/fsu022.

Farrell, E. D., L. Andersson, D. Bekkevold, et al. 2022. "A Baseline for the Genetic Stock Identification of Atlantic Herring, *Clupea harengus*, in ICES Divisions 6. A, 7." *Royal Society Open Science* 9, no. 9: 220453.

Fauchald, P. 2010. "Predatorprey Reversal: A Possible Mechanism for Ecosystem Hysteresis in the North Sea?" *Ecology* 91, no. 8: 2191–2197. https://doi.org/10.1890/09-1500.1.

Flores, J., J. Mena-Lorca, B. González-Yañez, and E. González-Olivares. 2007. "Consequences of Depensation in a Smiths Bioeconomic Model for Open-Access Fishery." *Proceedings of International Symposium on Mathematical and Computational Biology* 27: 219–232.

Frank, K. T., and D. Brickman. 2000. "Allee Effects and Compensatory Population Dynamics Within a Stock Complex." *Canadian Journal of Fisheries and Aquatic Sciences* 57, no. 3: 513–517. https://doi.org/10.1139/f00-024.

Froese, R., and D. Pauly. 2023. "FishBase." www.fishbase.org.

García-Carreras, B., S. Jennings, and W. J. F. Le Quesne. 2015. "Predicting Reference Points and Associated Uncertainty From Life Histories for Risk and Status Assessment." *ICES Journal of Marine Science* 73, no. 2: 483–493. https://doi.org/10.1093/icesjms/fsv195.

Gilroy, J. J., T. Virzi, R. L. Boulton, and J. L. Lockwood. 2012. "Too Few Data and Not Enough Time: Approaches to Detecting Allee Effects in Threatened Species." *Conservation Letters* 5, no. 4: 313–322. https://doi.org/10.1111/j.1755-263x.2012.00245.x.

Golden, A. S., B. van Poorten, and O. P. Jensen. 2022. "Focusing on What Matters Most: Evaluating Multiple Challenges to Stability in Recreational Fisheries." *Fish and Fisheries* 23, no. 6: 1418–1438. https://doi.org/10.1111/faf.12697.

González Herraiz, I., Y. Vila, M. Cardinale, et al. 2023. "First Maximum Sustainable Yield Advice for the *Nephrops norvegicus* Stocks of the Northwest Iberian Coast Using Stochastic Surplus Production Model in Continuous Time (SPiCT)." *Frontiers in Marine Science* 10: 1062078. https://doi.org/10.3389/fmars.2023.1062078.

Gonzalez-Duran, E., A. Hernandez-Flores, J. Carlos Seijo, A. Cuevas-Jimenez, and A. Moreno-Enriquez. 2018. "Bioeconomics of the Allee Effect in Fisheries Targeting Sedentary Resources." *ICES Journal of Marine Science* 75, no. 4: 1362–1373. https://doi.org/10.1093/icesjms/fsy018.

Griffiths, C. A., H. Winker, V. Bartolino, H. Wennhage, A. Orio, and M. Cardinale. 2024. "Including Older Fish in Fisheries Management: A New Age-Based indicator and Reference Point for Exploited Fish Stocks." *Fish and Fisheries* 25, no. 1: 18–37. https://doi.org/10.1111/faf.12789.

Gurney, W. S. C., P. J. Bacon, E. McKenzie, et al. 2010. "Form and Uncertainty in Stock-Recruitment Relations: Observations and Implications for Atlantic Salmon (*Salmo salar*) Management." *Canadian Journal of Fisheries and Aquatic Sciences* 67, no. 6: 1040–1055. https://doi.org/10.1139/F10-038.

Hassell, M. P. 1975. "Density-Dependence in Single-Species Populations." *Journal of Animal Ecology* 44, no. 1: 283–295.

Hedges, L. V., and I. Olkin. 1985. *Statistical Methods for Meta-Analysis*. Orlando, Florida, U.S.A.: Academic Press.

Hemeon, K. M., K. A. Ashton-Alcox, E. N. Powell, et al. 2020. "Novel Shell Stock-Recruitment Models for *Crassostrea virginica* as a Function of Regional Shell Effective Surface Area, a Missing Link for Sustainable Management." *Journal of Shellfish Research* 39, no. 3: 633–654. https://doi.org/10.2983/035.039.0310.

Hilborn, R., D. J. Hively, O. P. Jensen, and T. A. Branch. 2014. "The Dynamics of Fish Populations at Low Abundance and Prospects for Rebuilding and Recovery." *ICES Journal of Marine Science* 71, no. 8: 2141–2151. https://doi.org/10.1093/icesjms/fsu035.

Hüssy, K., C. M. Albertsen, J. Hemmer-Hansen, et al. 2022. "Where Do You Come From, Where Do You Go: Early Life Stage Drift and Migrations of Cod Inferred From Otolith Microchemistry and Genetic Population Assignment." *Canadian Journal of Fisheries and Aquatic Sciences* 79, no. 2: 300–313. https://doi.org/10.1139/cjfas-2020-0409.

Hüssy, K., H. Mosegaard, C. M. Albertsen, J. Hemmer-Hansen, E. E. Nielsen, and M. Eero. 2016. "Evaluation of Otolith Shape as a Tool for Stock Discrimination in Marine Fishes Using Baltic Sea Cod as a Case Study." *Fisheries Research* 174: 210–218. https://doi.org/10.1016/j.fishres.2015.10.010.

Hutchings, J. A. 2014. "Renaissance of a Caveat: Allee Effects in Marine Fish." *ICES Journal of Marine Science* 71, no. 8: 2152–2157. https://doi.org/10.1093/icesjms/fst179.

ICCAT (International Commission for the Conservation of Atlantic Tunas). 2019. "Compendium of the Management Recommendations and Resolutions Adopted by ICCAT for the Conservation of Atlantic Tunas and Tuna-Like Species." 417.

ICES. 2021a. "ICES Fisheries Management Reference Points for Category 1 and 2 Stocks (2021)." ICES Advice Technical Guidelines 16.4.3.1; p. 19. International Council for the Exploration of the Sea. https://doi.org/10.17895/ices.advice.7891.

ICES. 2021b. "WKREF1." https://github.com/ices-eg/WKREF1.

ICES. 2022a. "Workshop on ICES Reference Points." *ICES Scientific Reports* 4: 70. https://doi.org/10.17895/ices.pub.9749.

ICES. 2022b. "Workshop on Stock Identification of West of Scotland Sea Cod (WK6aCodID; Outputs From 2021 Meeting)." *ICES Scientific Reports* 4: 24. https://doi.org/10.17895/ices.pub.10031.

ICES. 2023a. "Benchmark Workshop on Northern Shelf Cod Stocks (WKBCOD)." *ICES Scientific Reports* 5: 425. https://doi.org/10.17895/ices.pub.22591423.

ICES. 2023b. "ICES Stock Assessment Database." https://standardgraphs.ices.dk.

ICES. 2023c. "ICES Stock Information Database." https://sid.ices.dk.

ICES. 2024. "Benchmark Workshop on Baltic Pelagic Stocks (WKBBALTPEL)." *ICES Scientific Reports* 5, no. 47: 350. https://doi.org/10.17895/ICES.PUB.23216492.

Iles, T. C. 1994. "A Review of Stock-Recruitment Relationships With Reference to Flatfish Populations." *Netherlands Journal of Sea Research* 32, no. 3: 399–420. https://doi.org/10.1016/0077-7579(94)90017-5.

Keith, D. M., and J. A. Hutchings. 2012. "Population Dynamics of Marine Fishes at Low Abundance." *Canadian Journal of Fisheries and Aquatic Sciences* 69, no. 7: 1150–1163. https://doi.org/10.1139/F2012-055.

Kell, L. T., I. Mosqueira, P. Grosjean, et al. 2007. "FLR: An Open-Source Framework for the Evaluation and Development of Management Strategies." *ICES Journal of Marine Science* 64: 640–646.

Kolody, D. S., J. P. Eveson, A. L. Preece, C. R. Davies, and R. M. Hillary. 2019. "Recruitment in Tuna RFMO Stock Assessment and Management: A Review of Current Approaches and Challenges." *Fisheries Research* 217: 217–234. https://doi.org/10.1016/j.fishres.2018.11.031.

Kramer, A. M., B. Dennis, A. M. Liebhold, and J. M. Drake. 2009. "The Evidence for Allee Effects." *Population Ecology* 51: 341–354. https://api.semanticscholar.org/CorpusID:36920194.

Kristensen, K., A. Nielsen, C. W. Berg, H. Skaug, and B. M. Bell. 2016. "TMB: Automatic Differentiation and Laplace Approximation." *Journal of Statistical Software* 70, no. 5: 1–21. https://doi.org/10.18637/jss.v070.i05.

Langangen, O., E. Edeline, J. Ohlberger, et al. 2011. "Six Decades of Pike and Perch Population Dynamics in Windermere." *Fisheries Research* 109, no. 1: 131–139. https://doi.org/10.1016/j.fishres.2011.01.029.

Li, N., and A.-A. Yakubu. 2012. "A Juvenile-Adult Discrete-Time Production Model Of Exploited Fishery Systems." *Natural Resource Modeling* 25, no. 2: 273–324. https://doi.org/10.1111/j.1939-7445.2011.00107.x.

Liermann, M., and R. Hilborn. 2001. "Depensation: Evidence, Models and Implications." *Fish and Fisheries* 2, no. 1: 33–58. https://doi.org/10.1046/j.1467-2979.2001.00029.x.

Liermann, M., and R. Hilborn. 1997. "Depensation in Fish Stocks: A Hierarchic Bayesian Meta-Analysis." *Canadian Journal of Fisheries and Aquatic Sciences* 54, no. 9: 1976–1984. https://doi.org/10.1139/f97-105.

Mandel, M. 2013. "Simulation-Based Confidence Intervals for Functions With Complicated Derivatives." *American Statistician* 67, no. 2: 76–81. https://doi.org/10.1080/00031305.2013.783880.

Marks, R., S. A. Hesp, A. Denham, N. R. Loneragan, D. Johnston, and N. Hall. 2021. "Factors Influencing the Dynamics of a Collapsed Blue Swimmer Crab (*Portunus armatus*) Population and Its Lack of Recovery." *Fisheries Research* 242: 106035. https://doi.org/10.1016/j.fishres.2021.106035.

Maroto, J. M., and M. Moran. 2014. "Detecting the Presence of Depensation in Collapsed Fisheries: The Case of the Northern Cod Stock." *Ecological Economics* 97: 101–109. https://doi.org/10.1016/j.ecolecon.2013.11.006.

Marshall, C. T., C. L. Needle, A. Thorsen, O. S. Kjesbu, and N. A. Yaragina. 2006. "Systematic bias in Estimates of Reproductive Potential of an Atlantic Cod (*Gadus morhua*) Stock: Implications for Stock-Recruit Theory and Management." *Canadian Journal of Fisheries and Aquatic Sciences* 63, no. 5: 980–994. https://doi.org/10.1139/F05-270.

McCarthy, M. A. 1997. "The Allee Effect, Finding Mates and Theoretical Models." *Ecological Modelling* 103, no. 1: 99–102. https://doi.org/10.1016/S0304-3800(97)00104-X.

Methot, R. D., and C. R. Wetzel. 2013. "Stock Synthesis: A Biological and Statistical Framework for Fish Stock Assessment and Fishery Management." *Fisheries Research* 142: 86–99. https://doi.org/10.1016/j.fishres.2012.10.012.

Millar, C., S. Large, A. Magnusson, and C. Pinto. 2022. "icesSAG: Stock Assessment Graphs Database Web Services." https://CRAN.R-project.org/package=icesSAG.

Morales-Bojórquez, E., and M. Nevárez-Martínez. 2005. "Spawner-Recruit Patterns and Investigation of Allee Effect in Pacific Sardine (*Sardinops sagax*) in the Gulf of California, Mexico." *California Cooperative Oceanic Fisheries Investigations Reports* 46: 161–174.

Myers, R. A., N. J. Barrowman, J. A. Hutchings, and A. A. Rosenberg. 1995. "Population Dynamics of Exploited Fish Stocks at Low Population Levels." *Science* 269, no. 5227: 1106–1108. https://doi.org/10.1126/science.269.5227.1106.

Nash, R. D. M., M. Dickey-Collas, and L. T. Kell. 2009. "Stock and Recruitment in North Sea Herring *Clupea harengus*; compensation and Depensation in the Population Dynamics." *Fisheries Research* 95, no. 1: 88–97. https://doi.org/10.1016/j.fishres.2008.08.003.

Needle, C. L. 2001. "Recruitment Models: Diagnosis and Prognosis." *Reviews in Fish Biology and Fisheries* 11, no. 2: 95–111. https://doi.org/10.1023/A:1015208017674.

Neuenhoff, R. D., D. P. Swain, S. P. Cox, et al. 2019. "Continued Decline of a Collapsed Population of Atlantic Cod (*Gadus morhua*) due to Predation-Driven Allee Effects." *Canadian Journal of Fisheries and Aquatic Sciences* 76, no. 1: 168–184. https://doi.org/10.1139/cjfas-2017-0190.

New Zealand Ministry of Fisheries. 2011. "Operational Guidelines for New Zealand's Harvest Strategy Standard." 80.

Nielsen, A., and C. W. Berg. 2014. "Estimation of Time-Varying Selectivity in Stock Assessments Using State-Space Models." *Fisheries Research* 158: 96–101. https://doi.org/10.1016/j.fishres.2014.01.014.

NRC (National Research Council). 2014. *Evaluating the Effectiveness of Fish Stock Rebuilding Plans in the United States*. Washington, DC: National Academies Press. https://doi.org/10.17226/18488.

Okamoto, D. K., R. J. Schmitt, S. J. Holbrook, and D. C. Reed. 2012. "Fluctuations in Food Supply Drive Recruitment Variation in a Marine Fish." *Proceedings of the Royal Society B: Biological Sciences* 279, no. 1747: 4542–4550. https://doi.org/10.1098/rspb.2012.1862.

Olsen, E. M., G. Ottersen, M. Llope, K.-S. Chan, G. Beaugrand, and N. C. Stenseth. 2010. "Spawning Stock and Recruitment in North Sea Cod Shaped by Food and Climate." *Proceedings of the Royal Society B: Biological Sciences* 278, no. 1705: 504–510. https://doi.org/10.1098/rspb.2010.1465.

Perala, T., J. A. Hutchings, and A. Kuparinen. 2022. "Allee Effects and the Allee-Effect Zone in Northwest Atlantic Cod." *Biology Letters* 18, no. 2: 20210439. https://doi.org/10.1098/rsbl.2021.0439.

Perala, T., and A. Kuparinen. 2017. "Detection of Allee Effects in Marine Fishes: Analytical Biases Generated by Data Availability and Model Selection." *Proceedings Of the Royal Society B-Biological Sciences* 284, no. 1861: 20171284. https://doi.org/10.1098/rspb.2017.1284.

Perälä, T., D. P. Swain, and A. Kuparinen. 2017. "Examining Nonstationarity in the Recruitment Dynamics of Fishes Using Bayesian Change Point Analysis." *Canadian Journal of Fisheries and Aquatic Sciences* 74, no. 5: 751–765. https://doi.org/10.1139/cjfas-2016-0177.

Pfister, C., and A. Bradbury. 1996. "Harvesting Red Sea Urchins: Recent Effects and Future Predictions." *Ecological Applications* 6, no. 1: 298–310. https://doi.org/10.2307/2269573.

Prince, J. D. 2013. "Cost-Benefit Analysis of Alternative Techniques for Rehabilitating Abalone Reefs Depleted by Abalone Viral Ganglioneuritis." *Reviews in Fisheries Science* 21, no. 3–4: 375–387. https://doi.org/10.1080/10641262.2013.837300.

Purchase, C. F., D. J. Hasselman, and L. K. Weir. 2007. "Relationship Between Fertilization Success and the Number of Milt Donors in Rainbow Smelt *Osmerus mordax* (Mitchell): Implications for Population Growth Rates." *Journal of Fish Biology* 70, no. 3: 934–946. https://doi.org/10.1111/j.1095-8649.2007.01356.x.

Quinn, J. F., S. R. Wing, and L. W. Botsford. 1993. "Harvest Refugia In Marine Invertebrate Fisheries: Models And Applications To the Red-Sea Urchin, *Strongylocentrotus franciscanus*." *American Zoologist* 33, no. 6: 537–550.

Quinn, T. P., C. J. Cunningham, J. Randall, and R. Hilborn. 2014. "Can Intense Predation by Bears Exert a Depensatory Effect on Recruitment in a Pacific Salmon Population?" *Oecologia* 176, no. 2: 445–456. https://doi.org/10.1007/s00442-014-3043-2.

RAM Legacy Stock Assessment Database. 2024. "RAM Legacy Stock Assessment Database v4.65 (Version v4.65)." Zenodo. https://doi.org/10.5281/zenodo.11995054.

Sass, G. G., Z. S. Feiner, and S. L. Shaw. 2021. "Empirical Evidence for Depensation in Freshwater Fisheries." *Fisheries* 46, no. 6: 266–276. https://doi.org/10.1002/fsh.10584.

Sguotti, C., S. A. Otto, R. Frelat, et al. 2019. "Catastrophic Dynamics Limit Atlantic Cod Recovery." *Proceedings of the Royal Society B: Biological Sciences* 286, no. 1898: 20182877. https://doi.org/10.1098/rspb.2018.2877.

Sharma, R., C. E. Porch, E. A. Babcock, M. N. Maunder, and A. E. Punt. 2019. "Recruitment: Theory, Estimation, and Application in Fishery Stock Assessment Models." *Fisheries Research* 217: 1–4. https://doi.org/10.1016/j.fishres.2019.03.015.

Shertzer, K. W., and M. H. Prager. 2007. "Delay in Fishery Management: Diminished Yield, Longer Rebuilding, and Increased Probability of Stock Collapse." *ICES Journal of Marine Science* 64, no. 1: 149–159. https://doi.org/10.1093/icesjms/fsl005.

Simpfendorfer, C. A. 2005. "Demographic Models: Life Tables, Matrix Models and Rebound Potential." In *Management Techniques for Elasmobranch Fisheries*, edited by J. A. Musick and R. Bonfil, 143–153. Rome: Food; Agriculture Organization of the United Nations.

Solari, A., J. MartinGonzalez, and C. Bas. 1997. "Stock and Recruitment in Baltic Cod (*Gadus morhua*): A New, Non-linear Approach." *ICES Journal of Marine Science* 54, no. 3: 427–443. https://doi.org/10.1006/jmsc.1996.0193.

Solari, A. P., M. T. G. Santamaria, M. F. Borges, et al. 2010. "On the Dynamics of *Sardina pilchardus*: Orbits of Stability and Environmental Forcing." *ICES Journal of Marine Science* 67, no. 8: 1565–1573. https://doi.org/10.1093/icesjms/fsq107.

Sugeno, M., and S. B. Munch. 2013a. "A Semiparametric Bayesian Approach to Estimating Maximum Reproductive Rates at Low Population Sizes." *Ecological Applications* 23, no. 4: 699–709. https://doi.org/10.1890/12-0453.1.

Sugeno, M., and S. B. Munch. 2013b. "A Semiparametric Bayesian Method for Detecting Allee Effects." *Ecology* 94, no. 5: 1196–1204. https://doi.org/10.1890/12-0454.1.

Swain, D. P., and H. P. Benoît. 2015. "Extreme Increases in Natural Mortality Prevent Recovery of Collapsed Fish Populations in a Northwest Atlantic Ecosystem." *Marine Ecology Progress Series* 519: 165–182.

Szuwalski, C. S., K. A. Vert-Pre, A. E. Punt, T. A. Branch, and R. Hilborn. 2014. "Examining Common Assumptions About Recruitment: A Meta-Analysis of Recruitment Dynamics for Worldwide Marine Fisheries." *Fish and Fisheries* 16, no. 4: 633–648. https://doi.org/10.1111/faf.12083.

Thygesen, U. H., C. M. Albertsen, C. W. Berg, K. Kristensen, and A. Nielsen. 2017. "Validation of Ecological State Space Models Using the Laplace Approximation." *Environmental and Ecological Statistics* 24, no. 2: 317–339. https://doi.org/10.1007/s10651-017-0372-4.

Tirronen, M., J. Depestele, and A. Kuparinen. 2023. "Can Regime Shifts in Reproduction Be Explained by Changing Climate and Food Availability?" *Frontiers in Marine Science* 10: 1167354. https://doi.org/10.3389/fmars.2023.1167354.

Tirronen, M., T. Perala, and A. Kuparinen. 2022. "Temporary Allee Effects Among Non-Stationary Recruitment Dynamics in Depleted Gadid and Flatfish Populations." *Fish and Fisheries* 23, no. 2: 392–406. https://doi.org/10.1111/faf.12623.

Todd, C. R., and M. Lintermans. 2015. "Who Do You Move? A Stochastic Population Model to Guide Translocation Strategies for an Endangered Freshwater Fish in South-Eastern Australia." *Ecological Modelling* 311: 63–72. https://doi.org/10.1016/j.ecolmodel.2015.05.001.

Trijoulet, V., C. M. Albertsen, K. Kristensen, C. M. Legault, T. J. Miller, and A. Nielsen. 2023. "Model Validation for Compositional Data in Stock Assessment Models: Calculating Residuals With Correct Properties." *Fisheries Research* 257: 106487. https://doi.org/10.1016/j.fishres.2022.106487.

Trijoulet, V., C. W. Berg, D. C. M. Miller, A. Nielsen, A. Rindorf, and C. M. Albertsen. 2022. "Turning Reference Points Inside out: Comparing MSY Reference Points Estimated Inside and Outside the Assessment Model." *ICES Journal of Marine Science* 79, no. 4: 1232–1244. https://doi.org/10.1093/icesjms/fsac047.

Trzcinski, M. K., R. Mohn, and W. D. Bowen. 2006. "Continued Decline of an Atlantic Cod Population: How Important Is Gray Seal Predation?" *Ecological Applications* 16, no. 6: 2276–2292.

United Nations. 1995. *Agreement for the Implementation of the Provisions of the United Nations Convention on the Law of the Sea of 10 December 1982 Relating to the Conservation and Management of Straddling Fish Stocks and Highly Migratory Fish Stocks*. New York: United Nations Conference on Straddling Fish Stocks; Highly Migratory Fish Stocks. A/CONF.164/37. https://www.un.org/oceancapacity/sites/www.un.org.oceancapacity/files/files/Projects/UNFSA/docs/unfsa_text-eng.pdf.

Vasconcellos, M. 2003. "An Analysis of Harvest Strategies and Information Needs in the Purse Seine Fishery for the Brazilian Sardine." *Fisheries Research* 59, no. 3: 363–378. https://doi.org/10.1016/S0165-7836(02)00026-7.

Vasilakopoulos, P., and C. T. Marshall. 2015. "Resilience and Tipping Points of an Exploited Fish Population Over Six Decades." *Global Change Biology* 21, no. 5: 1834–1847. https://doi.org/10.1111/gcb.12845.

Vergnon, R., Y.-J. Shin, and P. Cury. 2008. "Cultivation, Allee Effect and Resilience of Large Demersal Fish Populations." *Aquatic Living Resources* 21, no. 3, SI: 287–295. https://doi.org/10.1051/alr:2008042.

Winter, A.-M., A. Richter, and A. M. Eikeset. 2020. "Implications of Allee Effects for Fisheries Management in a Changing Climate: Evidence From Atlantic Cod." *Ecological Applications* 30, no. 1: e01994. https://doi.org/10.1002/eap.1994.

Winter, A.-M., N. Vasilyeva, and A. Vladimirov. 2023. "Spawner Weight and Ocean Temperature Drive Allee Effect Dynamics in Atlantic cod, *Gadus morhua*: Inherent and Emergent Density Regulation." *Biogeosciences* 20, no. 17: 3683–3716. https://doi.org/10.5194/bg-20-3683-2023.

Worm, B., and R. A. Myers. 2003. "Meta-Analysis of Cod-Shrimp Interactions Reveals Top-Down Control in Oceanic Food Webs." *Ecology* 84, no. 1: 162–173.

Yakubu, A.-A., N. Li, J. M. Conrad, and M.-L. Zeeman. 2011. "Constant Proportion Harvest Policies: Dynamic Implications in the Pacific Halibut and Atlantic Cod Fisheries." *Mathematical Biosciences* 232, no. 1: 66–77. https://doi.org/10.1016/j.mbs.2011.04.004.

Zebro, L. R., J. T. Mrnak, S. L. Shaw, S. R. Chipps, and G. G. Sass. 2022. "Density-Dependent and Environmental Influences on Juvenile Walleye *Sander vitreus* (Mitchill) Survivorship in Northern Wisconsin Lakes." *Fisheries Management and Ecology* 29, no. 6: 897–910. https://doi.org/10.1111/fme.12591.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.